

Clausal nominalization in Ka(g)wahíva

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Abstract Clausal nominalizations, which combine nominal properties externally with verbal properties internally, have often been understood as involving a nominal layer above the clausal content (Borsley & Kornfilt 2000; Kornfilt & Whitman 2011). Research based on this assumption proposes that the nominal layer consists of a determiner element D that nominalizes the clause structure rather than an *n* (or N). In this paper, I argue that clausal nominalizations do not always require a DP layer. Furthermore, I demonstrate that TP-nominalizations by *nP* do exist. I support both claims by examining nominalized complement clauses in the Tupí-Guaraní language Ka(g)wahíva. I show that complement clauses in this language are nominalized through *nP*, and do not have any additional nominal projections. Moreover, the clausal content consists solely of a TP. The first finding parallels clausal nominalization in Japanese and the Mongolic language Khalkha (Nasu et al. 2024; Lim 2025), while the second finding has only been argued for the Bantu language Oshiwambo (Lee & Ndapo 2025).

Keywords: Nominalization, Clausal nominalization, Tupí-Guaraní, Ka(g)wahíva

1 Introduction

In clausal nominalization, the internal properties of the clause point to a verbal core (e.g., finiteness morphology), while its external properties are nominal (e.g., nominal distribution, case marking). A frequently discussed issue about this topic is how the clause can behave as a nominal externally. Currently, there is a general consensus that a nominal layer lies on top of the clausal content. However, there is less agreement regarding the specific structure of the nominal layer. Two main positions have emerged, as schematized in (1).

- (1) a. [_{DP} D [... clausal structure ...]] (The D-assumption of clausal nominalization)
b. [_{nP} *n* [... clausal structure ...]] (The *n*-assumption of clausal nominalization)

In particular, some argue that in clausal nominalization, a determiner head D merges directly with the clausal content, as posited in (1a); I refer to this as the *D-assumption of clausal nominalization*, or simply the *D-assumption*. This position is taken in several works (Davies & Dubinsky 1998; Borsley & Kornfilt 2000; Takahashi 2010; Miyagawa 2011; Pietraszko 2019; Alexiadou 2020; Iordăchioaia 2020; Hankamer & Mikkelsen 2021; Bochnak & Hanink 2022)(cf. Grimshaw (2000) for a theoretical issue with this approach). In contrast, other scholars argue that the clausal content can be merged with *n* (Nasu et al. 2024; Lim 2025; 2026), as in (1b); throughout the paper, I call it the *n-assumption of clausal nominalization*, or just the *n-assumption*.

Studies that support the *D-assumption* in (1a) typically consider data indicating the presence of a determiner in clausal nominalizations, and rule out the *n-assumption* in (1b) based on expectations that are not met if an *nP* is assumed to exist within the same

structure. On the other hand, proponents of the *n-assumption* argue that a determiner element is not possible, suggesting that no DP is projected in these structures, even though the clausal constituent is nominal in distribution and morphosyntax. This leads to the proposal that the nominal layer *nP* must be present, although its head is phonologically null.

Against this backdrop, I examine clausal nominalization – specifically, complement clauses – in the Tupí-Guaraní language Ka(g)wahíva from South America. Like several other languages from South America (Comrie & Estrada-Fernández 2012; Zariquiey et al. 2019), Ka(g)wahíva requires nominalization of dependent clauses. Furthermore, Ka(g)wahíva presents a convenient set of properties in which to examine the nominal structure of clausal nominalization and test them against the predictions of the *D-assumption* and *n-assumption*. Notably, it has overt realization of the *nP* head through an enclitic, =*a*; overt exponence of elements within the determiner category; and it permits bare *nPs* (or small nominals, to borrow this terminology from Pereltsvaig (2006)).

Therefore, if Ka(g)wahíva clausal nominalizations follow the *D-assumption* structure, then we expect these structures to exhibit no enclitic =*a*, and exhibit the Ka(g)wahíva determiners. In contrast, if the language follows the *n-assumption* structure in clausal nominalizations, then these structures should exhibit =*a*, but not a determiner (overt or covert). As we will see, concerning the nominal structure within nominalized complement clauses, I argue that these structures are bare *nPs* and do not project a DP (or any higher nominal projection, such as PossP). As a result, Ka(g)wahíva clausal nominalizations support the *n-assumption* structure in clausal nominalizations, as in (1b).

This paper also addresses another important issue regarding clausal nominalization: the relationship between the clause size and the type of nominalizer used. Recent proposals indicate that the type of nominalizer determines how much structure it can embed (Alexiadou 2020; Iordăchioaia 2020; Lim 2025; 2026). This paper considers the claim that TP nominalizations are possible when the head D is involved, but not when using *n* (at least not when a DP layer is missing). I argue that this position is too restrictive for Ka(g)wahíva. I demonstrate that complement clauses are truncated structures that project up to TP. Thus, combined with the finding that the nominal structure in clausal nominalization only consists of *nP*, it leads to the conclusion that Ka(g)wahíva complement clauses are TP-embedding *nPs*.

If correct, these results add to recent proposals that nominalizations do not always require a DP layer, as also found in Japanese and the Mongolic language Khalkha (Nasu et al. 2024; Lim 2025), besides contributing to the only existing proposal that *nPs* may embed TPs without further nominal projections, as in the Bantu language Oshiwambo (Lee & Ndapo 2025).

This paper is organized as follows. After presenting more information about Ka(g)wahíva ecology and data methodology in §2 and briefly introducing the complement clause in §3, I present the properties of Ka(g)wahíva clause that will serve as a background to assess the clause-like nature of nominalized clausal complements, including clause structure 4.1 and verbal morphology 4.3. After that, I show that most, though not all, clausal properties are present in the complement clause in §5. This result will allow me to posit that these structures are truncated TP embeddings. Following that, I provide background on the properties of Ka(g)wahíva nominals. This section will be a more extensive discussion, since it will also motivate key claims about the structure of nominals in the language, chief among them the claim that there is no covert DP in §6. The next section discusses the nominal properties of complement clauses §7 to support the claim that nominalization

in complement clauses only involves an *nP* projection, and no DP. Finally, in Section 8, I explore the implications of this work and conclude in §9.

Before proceeding to the next section, I would like to emphasize that the amount of data and detail presented here is warranted. Ka(g)wahíva (and Tupían languages, more generally) has rarely been the subject of systematic studies on the nominalization of embedded clauses within a formal framework, despite the pervasiveness of nominalization in these languages. The notable exceptions include previous research on Ka(g)wahíva itself (dos Santos 2024b: in particular, Chapter 5 on nominalized relative clauses) and its sister language Tenetehára (Camargos & Castro 2013). More generally, some languages from distantly related branches within the broader Tupían family have been examined to explore the finiteness properties in dependent clauses as compared to root clauses, both in the Tuparí branch (Singerman 2021; Nogueira 2019: for Tuparí and Wayoro), and in the Arikém branch (Rocha da Silva 2016; Nogueira 2019; Vivanco 2021: for the single family member Karitiana).

2 Language ecology and data

Kagwahíva, or Kawahíva (in lects lacking the phoneme /gw/), is a critically endangered Amazonian language belonging to the Tupí-Guaraní language family. It is spoken by approximately 560 individuals out of a population of some 1060 individuals in total (dos Santos 2024a).

There are eight ethnic Ka(g)wahíva lects, often referred to by the names of the communities where they are spoken. In the state of Amazonas, the languages include Jiahui [ISO: pah], Júma [ISO: jua], Parintintin [ISO: pah], and Tenharin [ISO: pah]. In the state of Mato Grosso, there is Piripkura, and in the state of Rondônia, the languages spoken are Amondawa [ISO: adw], Karipuna [ISO: kuq], and Jupaú [ISO: urz] (also known by the exonym Uru Eu Wau Wau).

Unless otherwise indicated, the core set of examples provided in this paper is based on the author's fieldwork with native speakers of the Júma and Jupaú lects since 2019. These examples draw on annotated texts from spontaneous speech, primarily personal stories and traditional narratives. Additionally, targeted elicitation with speakers was conducted to confirm preliminary assumptions based on positive evidence from the texts.¹

¹ The particular methods employed are as follows: i) Recording, transcription, and translation of Ka(g)wahíva spontaneous speech; ii) Translation from Portuguese to Ka(g)wahíva; iii) Well-formedness judgments of Ka(g)wahíva utterances; iv) Comparison of the well-formedness of two or more Ka(g)wahíva utterances; v) Acceptability judgments of Ka(g)wahíva utterances in a constructed context; and vi) Description of visual stimuli in Ka(g)wahíva.

The recordings made using the method in (i) form a corpus of high-quality, carefully annotated audiovisual recordings of Ka(g)wahíva verbal art and conversational texts, including personal histories, traditional oral narratives, procedural descriptions, songs, and staged conversations. The annotation tasks in (i) followed a workflow that involves text segmentation, text respeaking (adapted from Basic Oral Language Documentation (BOLD) transcription method outlined in Reiman (2010), and text annotation.

The elicitation interview tasks described in (iii-vi) were primarily conducted with bilingual speakers in Ka(g)wahíva and Portuguese. For Júma, interviews were conducted with Mandéi Juma. For Jupaú, interviews were conducted with the Jupaú speakers Awip (Davi) and Puré.

All interviews were conducted with one speaker at a time. Translation tasks (ii) were used to obtain lexical, syntactic, and semantic information about Ka(g)wahíva. The tasks in (iii-iv) aimed to capture aspects of Ka(g)wahíva syntax. To indicate combinations that are not possible in the language, I marked sentences that were judged ill-formed in any context with an asterisk (*). The tasks in (v-vi) aimed to capture aspects of Ka(g)wahíva meaning that place constraints on the acceptability of well-formed utterances in certain situations. Utterances that are well-formed but not felicitous in a particular context are presented with a pound sign (#). During elicitation interviews, utterance contexts were constructed and supplied in Portuguese

This data is part of the online Ka(g)wahíva Language Documentation Archive, hosted at the California Language Archive (CLA); this is an accessible collection of audiovisual recordings of several Ka(g)wahíva lects.²

Work on Ka(g)wahíva grammar is still limited, but there have been recent advances on Ka(g)wahíva clause structure and verbal agreement (dos Santos 2023; 2025a; b). A recent grammar sketch is also available (dos Santos 2024b). The current paper draws on the current understanding of these two properties of Ka(g)wahíva.

In the following section, I briefly introduce the complement clause in Ka(g)wahíva.

3 The structure of interest

In almost all Tupí-Guaraní languages, including Ka(g)wahíva, nominalization is the strategy for embedding. Consider the notional complement clause within brackets in (2) below, from Ka(g)wahíva.

- (2) Okwaham ki Boreá-hěa amãna ikyre'yma ko'emamẽ]
 O-kwaham (ki) Boreá = hěa [amãna i-kyr-e'ym = a ko'emamẽ]
 3.A-know PST B. = 3SG.FEM rain MVD.OBJ-rain-NOM.NEG = NMLZ tomorrow
 ko.
 (ko)
 REALIS
 'Boreá thinks that it will not rain tomorrow.' (Júma: Elicit)

We will see that complement clauses like (2) exhibit the profile of nominals while also having a clause-like internal structure. For instance, they exhibit nominal negation (via *-e'ym*) and nominal tense (not shown in this example), in addition to appearing in typical nominal positions, like verbal objects; See Singerman (2018: on negation as an exclusively nominal category in the distantly related language Tuparí) and Tonhauser (2006; 2007) and Thomas (2014: on nominal temporal morphology in the sister languages Paraguayan Guaraní and Mbyá). Furthermore, these structures possess all the properties that matrix clauses exhibit up to TP. For instance, the clause-like syntax is evinced via adverbial modification in (2).

All in all, Ka(g)wahíva complement clauses can be best described as truncated TPs embedded under a nominal projection.

A note is in order regarding the English translations of Ka(g)wahíva complement clauses. Given the claim in section 5.1.4 that these structures are TP-embedded nominalizations in Ka(g)wahíva, I have tried to provide gerund translations for these examples whenever that was possible (e.g., 'You saw the jaguar killing a tapir', from (20) below), on a par with the idea that English gerunds are TPs. When this choice was not available, I gave translations with a finite CP, as in the *that*-clause in (2). These choices also reflect speakers' translations of Ka(g)wahíva into the target language, Portuguese, during elicitation tasks.

or supplied visually using picture stimuli, followed by a Ka(g)wahíva utterance intended to be spoken in that context.

² The collection is available through the following link: <http://dx.doi.org/doi:10.7297/X2P26W9H>

4 Clausal-like properties in Ka(g)wahíva

4.1 Clause structure

The clause-initial position in Ka(g)wahíva hosts a single element at a time. This initial element can be a verb (3a), an interrogative phrase (3b), a focused phrase (3c), or a topic phrase (3d).³

- (3) a. **Vepiang** po txi jirembiriko-ga Kanindé-gã nãhẽ.
V-epiang po txi ji = rembiriko = ga Kanindé = gã nãhẽ.
3.A-see IRR FUT 1SG = partner = 3SG.MASC Kanindé = 3.PL PROSP
 ‘My husband might see the Kanindé NGO group.’ (Júma: Elicit)
- b. **Mãngã** po jiruvyra vepiang rimba’e?
Mãngã po ji = ruvyra v-epiang rimba’e?
who IRR 1SG.POSS = uncle 3.A-see DISTANT.PST
 ‘Who did my uncle see back in the olden times?’ (Júma: Text)
- c. **Pirapetxinguhua** te’i ki hẽa opyhy-pyhy ko.
Pirapetxinguhua te’i ki hẽa o-pyhy~pyhy ko.
pirapetxinguhua.fish only PST 3SG.FEM 3.A-catch~RED REALIS
 ‘It was only *pirapetxinguhua* fish that she caught.’ (Júma: Text)
- d. **Takãi Kajuvi** ki, tapy’ynha opurun garehe.
Takãi.Kajuvi ki, tapy’ynha o-purun ga = rehe.
Takãi.Kajuvi PST non-indigenous 3.A-hit 3SG.MASC = at
 ‘(As for) Takãi Kajuvi, the non-Indigenous person ran over him.’ (Júma: Text)

Moreover, the first initial constituent always precedes the mood and tense particles; when the two are present, the particles follow the strict order Mood-Tense.⁴ In fact, no more than one initial element is allowed before the particles, no matter the linear order among the initial constituents, as in (4a).

- (4) a. *Vepiang Kanindé-gã po txi jirembiriko-ga.
 V-epiang Kanindé = gã **po txi** ji = rembiriko = ga.
 3.A-see Kanindé = PL IRR FUT 1SG.POSS = partner = MASC
 Intended: ‘It is Kanindé that may see my husband.’ (Júma: Elicit)
- b. *Kanindé-gã vepiang po txi jirembiriko-ga.
 Kanindé = gã v-epiang **po txi** ji = rembiriko = ga.
 Kanindé = PL 3.A-see IRR FUT 1SG.POSS = partner = MASC
 Intended: ‘It is Kanindé that may see my husband.’ (Júma: Elicit)

Assuming a layered approach to clause structure, the representation of the matrix clause structure preliminarily suggests itself as in (5). The CP, MoodP, and TP projections are supported by the properties of matrix clauses mentioned above, namely a single slot generally occupied by information-structurally prominent phrases in clause-initial

³ The full list of abbreviations is provided at the end of the paper (see Section Abbreviations). Furthermore, Ka(g)wahíva examples are presented using the practical Ka(g)wahíva writing system, which is similar to the IPA with the following exceptions: <y> = [i], <’> = [ʔ], <kw> [k^w], <tx> = [tʃ], <ngw> = [ŋg^w], <gã> = [ŋã]. Additionally, a nasal(ized) vowel is indicated with a tilde.

⁴ A parallel pattern, where a single initial element occurs in the clause-initial position and is followed by two sets of particles with a strict ordering (CP and TP particles), is found in Tuparí matrix and embedded clauses (Singerman 2020). Unlike Tuparí, however, Ka(g)wahíva lacks the mood and tense particles in embedded clauses, as shown in this paper for complement clauses (see Section 5.1.2).

position, as well as the mood and tense-related particles rigidly ordered after that initial position. The other projections, VoiceP, *v*P, and VP, are accounted for under a layered approach to clausal structure, as developed in various schools of syntactic analysis (Kratzer 1996; Pylkkänen 2002; Harley 2013).

(5) CP – MOODP – TP – VoiceP – *v*P – VP

Excepting for the verb-initial orders, I assume that phrase-initial elements are derived via [$\bar{A}$] movement. As for the verb-initial order VSO of pragmatically neutral contexts (see example (3a), I follow the account in dos Santos (2025a), in which the Ka(g)wahíva VSO results from long head movement to Spec,CP triggered by a categorial feature [V] with the EPP property on C. Diagram (6) represents this idea.

(6)

It is not crucial for the main claim of this paper, however, that information-structurally prominent phrases surface in the initial position by means of [$\bar{A}$] movement, and verbs do so via long head movement. What matters for our purposes is that matrix clauses exhibit a dedicated position before the mood and tense particles (or TP and MoodP in clausal-structural terms), which hosts pragmatically prominent phrases or the verb, identified as the CP.

4.2 Tense and negation

In Ka(g)wahíva, the expression of tense and negation notoriously differs across clause types (matrix clause versus dependent clauses). I first present the matrix tense particles, since, to my knowledge, no cognate particles with this semantics are found in sister languages.

The matrix tense particles that will matter for the discussion are the past tense particle *ki* and the future tense particle *txi*. The general past tense particle *ki* indicates that the action occurs anytime before the utterance time. In that way, it is compatible with past temporal adverbials, such as *oi'i* (~ *oji'i*) ‘some time ago’.

- (7) a. Gara ki nde ereapo oi'i?
 Gara **ki** nde ere-apo oi'i?
 what PST 2SG 2SG.A-do some.time.ago
 ‘What did you do yesterday?’ (Júma: Elicit)
- b. Humaitá uvi ore juarinẽ, ki ore hoi kavyripe
 Humaitá uvi ore ju = a = rinẽ, **ki** ore ho-i kavyr = ipe
 Humaitá from 1.EXCL come = NMLZ = AFTER PST 1.EXCL go-AIS jungle = in
 ko.
 ko.
 REALIS
 ‘After we returned from Humaitá city, we went into the jungle.’ (Júma: Elicit)

The future tense particle *txi* indicates that the action will occur sometime after the utterance time. It is compatible with future temporal adverbials, including *koi'iramẽ* ‘later’, and adverbial clauses with a future reference time.

- (8) a. Gara po txi nde ereapo koi'iramê? 'What will you do later?' (Júma:
Gara po **txi** nde ere-apo koi'iramê?
what IRR FUT 2SG 2SG.A-do later
Elicit)
- b. Wesley-ga horamê ko'emamê, txi jie mitakwanha poi
Wesley = ga ho = ramê ko'emamê, **txi** jie mitakwanha po-i
Wesley = 3SG.MASC go = WHEN tomorrow FUT 1SG bread make-AIS
nãhê. 'When Wesley leaves tomorrow, I will make bread.' (Júma: Elicit)
nãhê.
POT

As for matrix clause negation, pragmatically neutral negation is done through a circumfix, *n(da)-...-i*.

- (9) Ndajirepiagi gãmõ.
Nda = ji = repiag-**i** gã = mõ.
NEG-1SG.B = see-NEG 3.PL = INDF.DET
'Nobody saw me.' (Júma: Elicit)

In contrast, in dependent clauses, the exponents of tense and negation are bound morphemes that are also shared with simple nominals. Starting with tense, dependent clause tense surfaces through the suffixes *-ver* 'past tense' and *-ram* in Júma (and *-ham* in Jupaú) 'future tense'. Focusing on the past tense, the example in (10) illustrates this suffix within the complement clause. This morphology also occurs in simple nominals (11)-(12), where it exhibits lexically conditioned allomorphy (*-ver* ~ *-kwer*).

- (10) Kwarupi emõmbe'u ga vuva-ga repiagavera
Kwarupi e-mõmbe'u [ga v-uva = ga repiag-(a)**ver** = a
today 2SG.IMP-tell 3SG.MASC 3SG.COR = father = 3SG.MASC see-NOM.PST = NMLZ
oi'i!
oi'i]!
yesterday
'Tell (them) today that he saw his father yesterday.' (Júma: Elicit)
- (11) Ay-**ver** = a
wood-NOM.PST = NMLZ
'Stick' (Also applicable to any piece of wood that is no longer part of a tree)
(Júma: Text and Elicit)
- (12) Ji = rembireko-**kwer** = a = hêa
1SG.POSS = partner-NOM.PST = NMLZ = 3SG.FEM
'My ex-wife.' (Common in speech)

The negation exponent in dependent clauses is the suffix *-e'ym* (in Jupaú, *-i'im*). The same morpheme appears in simple nominals, where it realizes the idiosyncratic negative meaning of privative.

- (13) a. Emõmbe'u gã mǎjioka rerure'ymavera!
E-mõmbe'u [gã mǎjioka rerur-**e'ym**-(a)ver = a]!
2SG.IMP-tell 3.PL manioc bring-NOM.NEG-NOM.PST-NMLZ
'Tell (them) that they did not bring manioc!' (Júma: Elicit)

- b. U'ie'yma
 U'i-e'ym = a
 flour-NOM.NEG = NMLZ
 'I ate) flourless (rice).' (Júma: Text)

The distribution of nominal tense and negation morphology across simple nominals and dependent clauses is shared across the Tupí-Guaraní family (Seki 2000; Magalhães 2007; Praça 2007; Thomas 2014; Estigarribia 2020: to cite the best-known languages in this regard, Kamaiurá, Guajá, Tapirapé, Mbyá, and Paraguayan Guaraní, respectively). In fact, in the distantly related Tuparí language, the default marking of negation is nominal (Singerman 2018).

Next, I expand this background section on clausal properties by presenting those associated with verbs in matrix clauses.

4.3 Verbal properties

In this section, we will see that the typical verb in Ka(g)wahíva is subjected to reduplication, can be modified by adverbial PPs and suffixes, and exhibits valency morphology. An additional piece of morphosyntax discussed is verbal agreement.

4.3.1 Reduplication and adverbial modification

Reduplication is considered a morphological process restricted to verbs across the family (Rose 2023). In Ka(g)wahíva, reduplication encodes pluractionality. We can see verbal reduplication in (14a), which indicates multiple participants in an action, and in (14b), which indicates an iterated action.

- (14) a. Arerety-areretyk ki gã ko.
 Are = rety~areretyk ki gã ko.
 1.EXCL.B = pull.down~RED PST 3.PL REALIS
 'They pulled us all down.'
- b. Anhãnga okutu-kutupam jurua.
 Anhãnga **o-kutu~kutu-pam** j-urua.
 anhãnga.spirit 3.A-pierce~RED-completely MVD.OBJ-mouth
 'It was the *Anhãnga* spirit that pierced (his) mouth (many times).' (Jupaú: Text)

Adverbial modification can be indicated by affixation and postpositional phrases. Among the former are: *-pav* 'completely', *-jipe* 'already', *-jav* 'again', *-tehe* 'unpurposefully', and *-ahy* 'forcefully', and in the form of temporal postpositional phrases, common phrases are *ko'emamẽ* 'tomorrow' and *ypyjivete* 'early morning'.⁵

4.3.2 Valency morphology: causatives, applicative, reflexive, and reciprocal

The morphological expression of valency-augmenting and valency-decreasing semantics will also be considered, as it is traditionally treated as verbal within the family. I start with valency-increasing morphemes.

⁵ Other adverbial suffixes include *-katu* 'well', *-uhu* 'a lot', *-ete* 'really', *-i* 'a bit'. However, these elements, but not the ones from above, are also shared with simple nominals; with simple nouns, they mean *-katu* 'good', *-uhu* 'big', *-ete* 'genuine', *-i* 'small'. Therefore, I consider these adverbial suffixes unreliable as diagnostics of verbal modification.

Ka(g)wahíva has the causative prefix *mõ-* ($\sim$ *mbo-* in oral roots). It forms transitive verbs from intransitive verbs by introducing an agent/causer subject; compare (15a) to (15b).

- (15) a. O-hin ji = mën = a = ga.
3.A-fall 1.SG = partner = NMLZ = 3.SG.MASC
'My husband fell.' (Juma: Elicit)
- b. O-**mbo**-hin kurumĩ = a = gã nh-ãpepo = a.
3.A-CAUS-fall kid = NMLZ = 3.PL MVD.OBJ-pan = NMLZ
'The kids dropped someone's pan.' (Juma: Elicit)

Ka(g)wahíva also has the applicative morpheme *-okan* ($\sim$ *-ukan*). The suffix *-okan* adds an argument with a goal or benefactive role to a transitive verb, lexical or derived. This can be seen by contrasting the examples in (16a) and (16b), the latter of which has the applicative suffix.

- (16) a. O-mbo-hin kurumĩ = gã j-ãpepoa.
3.A-CAUS-fall kid = PL MVD.OBJ-pan
'The kids dropped (someone's) pan.' (Elicit)
- b. O-mbo-hir-**ukan** hëa mbiara javatxinga pe.
3.A-CAUS-fall-APPL 3.SG.FEM meat dog for
'She dropped the meat for the dog.' (Elicit)

Reflexivity and reciprocity are also expressed morphologically on the verb. Reflexivity is indicated by the prefix *ji-*, as in (17a), to mark the identity between subject and object. Reciprocity is expressed by *jo-*, which marks mutual action between the subject individuals, as in (17b).

- (17) a. O-**ji**-pe'a javytxinga jawara uvi.
3.A-REFL-distance dog jaguar from
'The dog distanced itself from the jaguar.' (Jupaú: Elicit)
- b. O-**jo**-vepiang ki gã ko.
3.A-RECP-see PST 3.PL REALIS
'They met.' (Lit.: They saw each other) (Juma: Elicit)

In sum, the typical morphosyntactic properties of Ka(g)wahíva verbs are reduplication, adverbial modification, and valency-changing morphology.

Finally, I introduce verbal agreement.

4.3.3 Verbal agreement

Ka(g)wahíva distinguishes grammatical relations primarily through verbal agreement with core arguments. In matrix clauses, verbs agree with either the subject or object; the choice is based on the highest-ranking person feature in the Person Hierarchy $1 > 2 > 3$ (dos Santos 2023). The examples in (18a)-(18b) illustrate that in sentences involving 1st and 2nd person, it is 1st person arguments that are privileged for verbal agreement over 2nd person arguments, regardless of their grammatical roles.

- (18) a. Ahepia ki ji nde ko.
A-hepia ki **ji** nde ko.
1SG.A-see PST 1SG 2SG REALIS
'I saw you.' (Júma: Elicit)

- b. Jirepia ki nde ko.
 Ji = repia ki nde ko.
 1SG.B = see PST 2SG REALIS
 ‘You saw me.’ (Júma: Elicit)

Table 1 shows the full set of agreement markers. As in several other sister languages, subject agreement markers are truly inflectional, whereas object agreement markers are pronominal clitics (Jensen 1990; 1998; Meira & Drude 2013; Rose 2015; dos Santos 2023). Note that there is no agreement marker for 3rd person objects.

	SET A ‘SUBJECT AGREEMENT’	SET B ‘OBJECT AGREEMENT’
1st person singular	a-	ji =
2nd person singular	ere-	nde =
3rd person	v- ~ o-	–
1st person inclusive	txi-	nhãnde =
1st person exclusive	oro-	ore =
2nd person plural	pe-	pe =

Table 1: Ka(g)wahíva agreement markers.

In contrast, in dependent clauses, the lexical verb may only agree with an object, using one of the object markers. In local scenarios of 1->2, as in (19a), agreement yields to 2nd person; agreement with the subject is not allowed, as (19b) shows.

- (19) a. Okwaha ki ga ji nderepiaka.
 O-kwaha ki ga [ji nde = repiak = a].
 3.A-know PST 3SG.MASC 1SG 2SG.B = see = NMLZ
 ‘He knew I saw you.’ (Júma: Elicit)
- b. *Okwaha ki ga ji nde ahepiaka.
 O-kwaha ki ga [ji nde a-hepiak = a].
 3.A-know PST 3SG.MASC 1SG 2SG 1SG.A-see = NMLZ
 Intended: ‘He knew I saw you.’ (Júma: Elicit)

This pattern of object agreement in dependent clauses is further evidenced in non-local scenarios (i.e., 3 > 3). Recall from Table 1 that Ka(g)wahíva does not exhibit agreement with 3rd person objects, while agreement with 3rd person subjects renders the prefix *o-* on the lexical verb. In dependent clauses with third person (3->3), the presence of the subject marker prefix *o-* creates an ungrammatical structure, as shown by (20).

- (20) a. *Erepia ki nde jawara tapi’ira ojuka ko.
 Ere-pia ki nde [jawara tapi’ira o-juka = a] ko.
 2SG.A-see PST 2SG jaguar tapir 3.A-kill = NMLZ REALIS
 Intended: ‘You saw the jaguar killing a tapir.’ (Júma: Elicit)
- b. Erepia ki nde jawara tapi’ira juka ko.
 Ere-pia ki nde [jawara tapi’ira juka = a] ko.
 2SG.A-see PST 2SG jaguar tapir kill = NMLZ REALIS
 ‘You saw the jaguar killing a tapir.’ (Júma: Elicit)

The Ka(g)wahíva agreement system is analyzed in dos Santos (2025a; b). Assuming the standard view that agreement is the result of a probe-goal relationship mediated through Agree (Chomsky (2001)), dos Santos argues that these differences in agreement require the view where subject and object agreement markers reside in different loci, and each probe

has a different featural specification in Ka(g)wahíva. In particular, subject markers are due to a probe relativized to [SPKR-PART- Φ] on C, while object markers result from a flat [Φ]-probe on *v*. The combination of this idea with the clause structure proposed in the previous sections yields the representation of matrix clauses with two probes in (21)

- (21) [CP C_[SPKR-PART-Φ]_[V] MoodP TP [VoiceP EA Voice νP ν_[Φ] VP IA V]]]]]]

In this account, because only the C head bears a probe that c-commands (and, therefore, may copy features from) both core arguments, person hierarchy effects/subject agreement may only arise in the combination of two processes: the presence of the C head and the movement of the verb to CP to serve as a host for this morphology. Let us call these the Ka(g)wahíva Syntactic Conditions on Agreement:

- (22) Ka(g)wahíva Syntactic Conditions on Person Hierarchy Effects/Subject Agreement:
 a) Merging of a C-head with the relativized probe [SPKR-PART-Φ]
 b) Movement of the verb to CP

These two conditions straightforwardly capture the agreement facts in VSO clauses. In VSO clauses, as in example (23), the verb appears in clause-initial position – and by hypothesis undergoes movement to the CP level – and therefore bears agreement with the highest-ranking person feature in the Person Hierarchy $1 > 2 > 3$; if the subject is the 1st person, then subject agreement yields:

- (23) Ahepia ki ji nde pevo ko.
A-hepia ki ji nde pevo ko.
1SG.A-see PST 1SG 2SG there REALIS
'I saw you there.' (Júma: Elicit)

In contrast, in dependent clauses, as in (19b)-(20b), verbs are strictly in final position relative to core arguments, which is taken to indicate the absence of movement to CP. We also see that agreement is restricted to objects in these instances. Under the idea that features from the external argument are only available on C, verb movement to CP in VSO clauses, but not in SOV clauses, correctly predicts that VSO clauses may yield agreement to the external argument, but SOV clauses should not.⁶

In sum, Ka(g)wahíva subject agreement results from a probe that sits high in the clause structure, the CP level; a similar agreement system in the sister language Mbyá-Guaraní has recently received an analogous analysis (Thomas et al. 2025).

4.4 Interim summary

The characterization of Ka(g)wahíva matrix clauses and the verbal morphosyntax provides several properties for assessing how clause-like other structures are, such as complement clauses. That is, if we find all or only a subset of these properties, this strongly suggests that the other structures under examination are either structurally akin to matrix clauses or truncated at a certain height.

⁶ A reviewer expressed curiosity about subject agreement distribution in non-verb-initial clauses with DP extraction, as in examples (3b), (3c), and (3d). In these, we see that verbs are not initial, yet they appear with the agreement associated with the CP. Since the details of how subject agreement in these instances arises are beyond the scope of this paper, I will only refer the interested reader to dos Santos (2025a), which deals with the distribution of agreement across other clause types, including those in the non-verb-initial sentences.

For Ka(g)wahíva, we will see that complement clauses exhibit most of the properties reviewed here for matrix clauses, but lack some: thus, they have the morphosyntax that matrix clauses also show up to the TP, but entirely lack those associated with the CP, such as reality mood and subject agreement.

In turn, determining the size of complement clauses will later serve to investigate the different expectations that approaches to clausal nominalization, the *D-assumption* and *n-assumption*, make for complement clauses if they are TP-embeddings. In particular, under the *D-assumption* structure, TP is expected. In contrast, if these nominalizations involve the *n-assumption*, a TP structure is not expected.

5 Clause-like properties of complement clauses

In this section, I show that complement clauses may exhibit clausal material up to TP, but no further. This will lead me to posit that these clauses are truncated, projecting up to TP.

5.1 Amount of clause structure

5.1.1 Presence of TP in complement clauses

Complement clauses involve at least a TP. That tense is projected in embedded clauses is shown by the fact that they allow temporal postpositional phrases like *ko'emamẽ* in (24), which unambiguously modify the embedded verb.

- (24) Okwaham ki Boreá-hěa amãna ikyre'yma ko'emamẽ
 O-kwaham (ki) Boreá = hěa [amãna i-kyr-e'ym = a ko'emamẽ]
 3.A-know PST B. = 3SG.FEM rain MVD.OBJ-rain-NEG = NMLZ tomorrow
 ko.
 (ko)
 REALIS
 'Boreá thinks that it will not rain tomorrow.' (Júma: Elicit)

In fact, each clause may appear with a temporal adjunct that mismatches in tense interpretation, as in (25).

- (25) Kwarupi emõmbe'u ga vuva-ga repiagavera
 Kwarupi e-mõmbe'u [ga v-uva = ga repiag-(a)ver = a
 today 2SG.IMP-tell 3SG.MASC 3SG.COR = father = 3SG.MASC see-PST = NMLZ
 oi'i!
 oi'i!
 yesterday
 'Tell (them) today that he saw his father yesterday.' (Júma: Elicit)

A further piece of evidence that a TP is projected in complement clauses is evinced by the presence of the dependent nominal past tense marker *-ver* in examples like (26), showcasing long-distance extraction of the subject of the most embedded clause. In both dependent clauses, the most embedded verb and the intermediate verb appear with the dependent past tense marker *-ver*.⁷

⁷ A reviewer wonders if (26) does not characterize a violation of the Complex NP Constraint, a parallel of which is the English *Who did Mary tell [the rumor [that _ was stealing clothes]]? It does not, as Ka(g)wahíva structures like (26) cannot be assimilated into the profile of NP complement constructions. Unlike the English

- (26) Mǎngǎ ga omômbe'u nde epiakavera ipira imĩmavera?
 Mǎngǎ_i ga o-mômbe'u [nde epiak-**aver** = a [_i i-pira i-mĩm-**aver** = a]]?
 who 3SG.MASC 3.A-tell 2SG see-**PST** = NMLZ MVD.OBJ-clothes MVD.OBJ-hide-**PST** = NM
 'Who did he tell that you saw stealing clothes?' (Júma: Elicit)

Taken together, these facts suggest that Ka(g)wahíva complement clauses project TPs.⁸ Next, I present evidence that projections below TP in matrix clauses are also present in complement clauses.

5.1.2 Finiteness properties of complement clauses

Complement clauses exhibit several verbal properties (akin to English *-ing* gerunds) (Alexiadou 2020; Iordăchioaia 2020: among others). Beginning with their verbal properties, these nominalizations show adverbial modification in the form of verbal suffixes (27a)-b) and postpositional phrases (27c) (see also (24)-(25) from above).

- (27) a. Emômbe'u tapy'ynha-gǎ rurahyavera!
 E-mômbe'u [tapy'ynha = gǎ rur-**ahy**-(a)ver = a]!
 2SG.IMP-tell non-indigenous = 3.PL come-**FORCEFULLY**-NOM.PST = NMLZ
 'Tell (them) that the non-indigenous people came (on a motorcycle) crazily.'
 (Júma: Elicit)
- b. Emômbe'u gǎ mǎjioka reruratehevera!
 E-mômbe'u [gǎ mǎjioka rerur-**atehe**-ver = a]!
 2SG.A-tell 3.PL manioc bring-**IN.VAIN**-NOM.PST = NMLZ
 'Tell (them) they brought manioc for no reason.' (Júma: Elicit)
- c. Emômbe'u tapy'ynha-gǎ pira ipyhykavera orekavyra
 E-mômbe'u [tapy'ynha = gǎ pira i-pyhyk-(a)ver = a ore = kavyra
 1SG.A-tell non-indigenous = 3.PL fish MVD.OBJ-catch-NOM.PST = NMLZ 1.EXCL = land
 pyri!
 pyri]!
 near
 'Tell (them) that the non-indigenous people caught fish in our land!' (Júma: Elicit)

Other projections include voice morphology in the form of causative (28a), reflexive (28b), and reciprocal (28c), pluractional aspect in the form of reduplication (28d), which also exhibits applicative morphology, negation (28e), and tense morphology in the form of the temporal past suffix *-(a)ver* (all examples).

- (28) a. Emômbe'u ga mbahira imbovevavera!
 E-mômbe'u [ga mbahira i-**mbo**-vev-(a)ver = a]!
 2SG.IMP-tell 3SG.MASC light MVD.OBJ-**CAUS**-turn.off-NOM.PST = NMLZ
 'Tell (them) that he turned off the light!' (Júma: Elicit)

example, the Ka(g)wahíva one does not involve an overt noun (*claim*, *rumor*, *fact*, nor are the nominalized complements CPs, as the present section argues.

⁸ Other diagnostics to probe the presence of a TP are unavailable in Ka(g)wahíva. For instance, Lim (2025; 2026), probing the existence of TP in complement clauses in Khalkha, relies on scope evidence relating the interpretation of universally quantified subjects to negation, which he assumes is under TP; if universal quantifiers as subjects scope above negation, then it is because the subject has A-moved to TP. This diagnostic is not available in Ka(g)wahíva because the language shows no evidence that subjects move to TP.

- b. Emõmbe'u gã japoavera!
 E-mõmbe'u [gã j-apo-(a)ver = a]!
 2SG.IMP-tell 3.PL REFL-do-NOM.PST = NMLZ
 'Tell (them) that they hurt themselves!' (Júma: Elicit)
- c. Emõmbe'u gã jomõngita-mõngitavera!
 E-mõmbe'u [gã jo-mõngita-mõngita-ver = a]!
 2SG.IMP-tell 3.PL RECP-narrate.story-RED-NOM.PST = NMLZ
 'Tell (them) that they repeatedly narrated stories to each other!' (Júma: Elicit)
- d. Emõmbe'u Agui-ga mbiara mbohirukaravera javatxinga
 E-mõmbe'u [Agui = ga mbiara mbohir-**ukar**-(a)ver = a javatxinga
 2SG.IMP-tell A. = 3SG.MASC meat drop-APPL-NOM.PST = NMLZ dog
 pe!
 pe]!
 to
 'Tell (them) that Agui dropped meat for the dog!' (Júma: Elicit)
- e. Emõmbe'u gã mǎjioka rerure'ymavera!
 E-mõmbe'u [gã mǎjioka rerur-**e'ym**-(a)ver = a]!
 2SG.IMP-tell 3.PL manioc bring-NOM.NEG-NOM.PST-NMLZ
 'Tell (them) that they did not bring manioc!' (Júma: Elicit)

All these examples also illustrate that clausal complements have event participants (subjects and objects) and event readings.

In the following section, I show that clausal complements still lack several other properties of matrix clauses.

5.1.3 Absent finiteness properties and main clause phenomena in complement clauses

Complement clauses are structurally reduced from full clauses – they do not allow mood marking, evidentiality, some types of aspect, subject agreement, and clause-internal fronting, all associated with projections higher than TP. The initial piece of evidence for the reduced size of embedded clauses is their lack of proposition-marking elements, such as reality mood (29a) and propositional morphology, such as *-aha* 'unfortunately' (29b).

(29) No mood marking in clausal nominalizations

- a. *Arovia jie Puré-ga po tapi'ira juka.
 A-rovia jie [Puré = ga **po** tapi'ira juk = a]
 1SG.A-believe 1SG Puré = 3SG.MASC IRR tapir kill = NMLZ
 Intended: 'I believe that Puré may have killed a tapir.' (Júma: Elicit)
- b. *Arovia jie po Puré-ga tapi'ira juka.
 A-rovia jie [**po** Puré = ga tapi'ira juk = a]
 1SG.A-believe 1SG IRR Puré = 3SG.MASC tapir kill = NMLZ
 Intended: 'I believe that Puré may have killed a tapir.' (Júma: Elicit)

- (30) *Emõmbe'u gã mǎjioka eruraha!
 E-mõmbe'u [gã mǎjioka erur-**aha**]!
 2SG.IMP-tell 3PL manioc bring-UNFORTUNATELY
 Intended: 'Tell (them) they unfortunately brought manioc!' (Júma: Elicit)

Subject agreement, too, is ruled out from dependent clauses, as shown by the ungrammaticality of 3rd person subject agreement in the bracketed complement clause in (31).

- (31) *Erepia ki nde jawara tapi'ira ojuka ko.
 Ere-pia ki nde [jawara tapi'ira o-juka = a] ko.
 2SG.A-see PST 2SG jaguar tapir 3.A-kill = NMLZ REALIS
 Intended: 'You saw the jaguar killing a tapir.' (Júma: Elicit)

Additionally, these structures do not allow two other properties of matrix clauses: evidentiality and habitual and progressive aspect. The former category, in the form of suffixes like *auhu* 'seemingly' and particles, creates ill-formed structures in complement clauses (32).

- (32) a. *Emõmbe'u gã auhu mǎjioka reruravera!
 E-mõmbe'u [gã **auhu** mǎjioka rerur-(a)ver = a]!
 2SG.IMP-tell 3PL **SEEMINGLY** manioc bring-NOM.PST = NMLZ
 Intended: 'Tell that they seemingly brought manioc!' (Júma: Elicit)
 b. *Emõmbe'u avoipava ka'ia i'ue'yma mǎipo
 E-mõmbe'u [avoipava ka'ia i-'u-'ym = a **mǎipo**]
 2SG.IMP-tell ancestor monkey MVD.OBJ-eat-NOM.NEG = NMLZ N.WIT.PST
 Intended: 'Tell that the ancestors did not eat monkey in the olden times!' (Júma: Elicit)

A last piece of finiteness property from matrix clauses that is absent in complement clauses is the expression of two subcategories of aspect – habitual aspect, through the subject inflected auxiliary *ko* 'be', and progressive aspect, via inflected posture auxiliaries like *ten* 'sitting'. Recall from Section 5.1.2 that the pluractional aspect is nonetheless allowed in such structures.⁹

- (33) *Emõmbe'u jie matera 'ua itena!
 E-mõmbe'u [jie matera 'u = a **i-ten = a**]
 2SG.IMP-tell 1SG food eat = NMLZ 1SG.COR-sit = NMLZ
 'Tell (them) that I am eating food, sitting!' (Júma: Elicit)
 (34) *Emõmbe'u gã mǎjioka erua okovo!
 E-mõmbe'u gã [mǎjioka eru = a **o-ko-vo**]
 2SG.IMP-tell 3PL manioc bring = NMLZ 3.COR-be-DEPEND
 Intended: Tell that they habitually bring manioc!' (Júma: Elicit)

Some syntactic phenomena from main clauses are also lacking in complement clauses. Examples (35) show that complement clauses do not allow embedded questions. In elicitation, the strategy of questioning a phrase within the embedded clause is always rejected, as the examples below demonstrate.

- (35) a. *Akwaham jie mōrōmō pe jimitakwanha i-'u = a
 A-kwaham jie [**mōrōmō** pe ji = mitakwanha i-'u = a]
 1SG.A-know 1SG **which** 2PL 1SG = bread MVD.OBJ-eat = NMLZ
 Intended: 'I know which one of you ate my bread.' (Júma: Elicit)

⁹ A reviewer wonders why the choice for *-vo*, in *okovo* from example (34), and not *=a*. While confirmation that this combination with *=a* will have to wait, I suspect it should not be possible as habitual semantics is encoded via *eko* 'be' marked with *-vo*.

(39) TP – VoiceP – ν P – VP

In brief, the Ka(g)wahíva matrix and complement clauses exhibit asymmetries in the amount of structure they project. Matrix clauses project a CP and MoodP, which are missing in complement clauses.

In this way, Ka(g)wahíva complement clauses, as TP embeddings, are similar to TP embeddings in languages like Turkish (Kornfilt & Whitman 2011). The parallel becomes even more evident when nominalization is considered, as Turkish TP embeddings involve nominalization, which we will see later that Ka(g)wahíva also does. We will see, however, that a complete assimilation of the Ka(g)wahíva TP embedded nominalization into a pattern like Turkish is not possible: while Ka(g)wahíva subjects in TP embeddings do not pattern like possessors, subjects in Turkish do (the Turkish genitive subject).

Having motivated this claim, I will, in Section 7, argue that clausal nominalizations contain nominal structure. First, I provide a detailed characterization of nominals in the language.

6 Nominal-like properties in Ka(g)wahíva

In this section, I present the two key properties of Ka(g)wahíva nominals relevant for the discussion of nominalization in complement clauses. The first one is that Ka(g)wahíva nominals are the result of an acategorial root that undergoes syntactic derivation with the functional head n , spelled out as $=a$. Secondly, Ka(g)wahíva determiners only combine with n Ps whose referents are living humans (and God). Alongside the discussion of the first property, I also provide a brief comparison to the cognate $=a$ across the family and show that bare nouns are just bare n Ps, not covert DPs.

These properties will serve as a framework against which clause-sized nominals can be compared and probed for their nominal structure and whether they are amenable to the *D-assumption* or *n-assumption* of clausal nominalizations. A summary of what each approach to these structures predicts is given in Table 2.

	$n =_a$	determiner
D-assumption	✗	✓
n -assumption	✓	✗

Table 2: Properties of Ka(g)wahíva nominals under approaches to clausal nominalization.

6.1 The morpheme $=a$ in Ka(g)wahíva

A hallmark of Ka(g)wahíva nominals is the presence of the enclitic $=a$, as shown for a common noun in (40). Proper names, too, require the enclitic $=a$ (and a definite marker enclitic) (41a), unless they are used as a vocative, as in (41b).^{10,11}

- (40) *tajahua*
 tajahu = a
 peccary = NMLZ
 ‘Peccary’

¹⁰ Speakers may also drop the enclitic from Portuguese proper names.

¹¹ I briefly return to this distribution in vocatives in Section §6.5, footnote 21.

- (41) a. Aviva-ga pamẽ ga hoi, 'iko.
Aviv = a = ga pamẽ ga ho-i, 'i:ko.
Avip = NMLZ = 3SG.MASC together 3SG.MASC go-AIS DIR.EVID
 'He was together with Avip.' (Júma: Text)
- b. Avip, erun yhya oreve.
Avip, e-run yhya ore = ve.
Avip, 2SG.IMP water 1.EXCL = for
 'Avip, bring water for us.' (Jupaú: Overheard)

This distribution is further demonstrated in (42): the absence of =a on words in the syntactic position of a noun is ungrammatical, such as heads of possessive phrases and objects of a transitive clause.

- (42) a. javatxinga ra'yra
 javatxing = a ra'yr = *(a)
 dog = NMLZ offspring = NMLZ
 'Dog's puppy.' (Júma: Elicit)
- b. Ahepiang ki jie javatxinga ra'yra ko.
 A-hepiang ki jie javatxing = a ra'yr = *(a) ko.
 1SG.A-see PST 1SG dog = NMLZ offspring = NMLZ REALIS
 'I saw a dog's puppy.' (Júma: Elicit)

The formative =a also appears with members from other lexical classes, such as postpositions, number words (which are adverbial), and verbs, in which case the resulting combination distributes as a noun. This is demonstrated in examples in (43) for a postposition-based nominal (43a) and an adverbial number word (43b), which must bear the nominalizer as a complement of a postposition and as the subject of an intransitive verb, respectively.¹²

- (43) a. Pea uvi gã hoi.
Pe = *(a) uvi gã ho-i.
to = NMLZ from 3.PL go-AIS
 'From there they went.' (Text and Elicit)
- b. A'ipea inhãni ovovo.
A'ipe = *(a) i-nhãn-i o-vo-vo.
one = NMLZ MVD.OBJ-run-AIS 3.COR-go-DEPEND
 '(Then), one ran away and went.' (Text and Elicit)

As indicated, a pervasive use of =a is in combination with verbs, creating deverbal nouns, as in (44), or clause-sized nominals, as in (45); in the latter, the resulting nominal can serve as a complement to clause-embedding verbs, such as *hepiang* 'see' and *kwaham* 'know'.

- (44) a. Jiu'uavera
 Ji = u'u-(a)ver-a
 1SG.POSS = bite-NOM.PST-NMLZ
 'My bite.' (Júma: Text)

¹² Words that express nominal quantification in other languages, such as numerals and quantifiers, pattern as adjuncts in the language.

- b. 'aravera
 'ar-(a)ver = a
 be.born-NOM.PST = NMLZ
 'Birthday (lit. someone's falling).' (Common in speech)
- (45) a. Ahepia ji Puré-ga piauhua, ipyhyga.
 A-hepia ji [Puré = ga piauhua, i-pyhyg = a].
 1SG.A-see 1SG Puré = 3SG.MASC *piauhua*.fish MVD.OBJ-catch = NMLZ
 'I saw Puré catching a *piauhua* fish.' (Júma: Elicit)
- b. Akwaham ji Puré-ga piauhua, ipyhyga.
 A-kwaham ji [Puré = ga piauhua, i-pyhyg = a].
 1SG.A-know 1SG Puré = 3SG.MASC *piauhua*.fish MVD.OBJ-catch = NMLZ
 'I know Puré caught a *piauhua* fish.' (Júma: Elicit)

Next, I show that the distribution of cognate = *a* across the Tupí-Guaraní language family is consistent with that presented here.

6.2 =*a* across the Tupí-Guaraní family

Cabral (2001) provides a broad survey of = *a* within the Tupí-Guaraní family, according to which a set of languages has = *a* as a grammatically productive formative, while the other set comprises languages in which = *a* only shows as crystallized in some nominal forms.

In languages with productive = *a*, this formative generally patterns as shown for Ka(g)wahíva above. This fact can be demonstrated for Kamaiurá and Tapirapé, for which there is a systematic description of cognates of = *a* that allows a comparison to Ka(g)wahíva. I follow the original sources in notating this morpheme as a suffix, and thus set it off with a dash, rather than the equals sign as with the Ka(g)wahíva cognate.¹³

In Kamaiurá, according to Seki (2000), = *a* appears with simple nouns occurring as verbal arguments (46a), complements of postpositions (46b), and possessed nouns (46c). Furthermore, = *a* combines with the verb of clausal nominalizations, as in the notional

¹³ The other languages that Cabral (2001)'s survey indicates as having a grammatical -*a* include (several etnovarieties of) Ka(g)wahíva, Asuriní do Tocantins, Parakanã, Suruí (Aikewara), Tembê, Tupinambá, Guaraní Antigo, Avá-Canoeiro, Asuriní do Xingu, Turiwára, Anambê, Wajampí do Amaparí, Emérillon, and Jo'é.

Only for some of these languages has there been later grammatical work that provides a complete, or almost complete, characterization of -*a* with illustrative examples. In all of them, the cognate -*a* shows the same distribution as in Ka(g)wahíva, Kamaiurá, and Tapirapé. For instance, this is true for Asuriní do Xingu (Pereira 2009), Asuriní do Trocará (Vieira 1993), Avá-Canoeiro (Borges 2006), and Emérillon (Rose 2003), with some caveats for the last three: the distribution of -*a* cannot be shown for (complement) clauses in Asuriní do Trocará, as they only involve direct quotation; no data is presented for embedded clauses in Avá-Canoeiro. As for Emérillon, notional complement clauses do not show -*a*. This absence of -*a* can be explained in Emérillon as it has lost the nominalization strategy for dependent clauses; for instance, in Emérillon, but not in sister languages with that strategy, these structures can exhibit subject agreement, and there is no distinction in negation marking between matrix and dependent clauses.

Of the remaining languages in the list, most lack a comparable dataset for -*a*. These are: Guaraní Antigo, Jo'é, Parakanã, Tembê, Tupinambá, and Turiwára.

Finally, for Anambê, the only grammatical work on the language explicitly states that it has no -*a* (Julião 2005). This scenario also extends to one variety of the closely related language Wayampí: one variety does not have -*a* (Copin 2012), while the other variety apparently does, at least given its segmentation in simple nouns in (Copin 2026); however, no grammatical work exists on the function of this morpheme in this variety. The presence of the formative -*a* is also observed in the only accessible work for Suruí (Aikewara), Lopes (2014), as it contains examples with an -*a* segmented off simple nouns; however, discussion about its distribution is lacking in that source.

complement clause in (46d); the source draws on this distribution, and others, to diagnose the nominal status of clausal embeddings.¹⁴

- (46) a. Moĩ-a wararuwijaw-a o-u'u.
snake-N dog-N 3-bite
'The snake bit the dog.' (Seki 2000: 166, ex. 530)
- b. Kunu'um-a o-jae'o motawa-a rehe
boy-N 3-cry food-N for
'The boy is crying because of food.' (Seki 2000: 58, ex. 11)
- c. Āaaa je = rajyr-a te = he = pa.
AH! 1SG = daughter-N FOC = VIS = MS
'Ah! It's my daughter indeed.' (Seki 2000: 346, ex. 1315e)
- d. A-kwahaw = ete = rak [mokaw-a mepy-taw-er-a].
1SG-know = ACTUAL = ATT gun-N buy-NMLZ-PST-N
'I know that (he) bought a gun.' (Seki 2000: 171, ex. 556)

In contrast, =a does not appear in vocatives (contrast (47a)-(47b) and forms that would otherwise be simple nouns with this formative distribute as verbal predicates, being marked, e.g., for negation identically to how verbs are (47c)-(47d).

- (47) a. A-ha kor = a'e = wa api
1SG-go FS = N.INT = MS VOC
'I'll go, Uncle.' (Seki 2000: 60, ex. 20)
- b. 'Aŋ a'ikĩ api-a rur-i a'e
now ATEN uncle-N come-AIS N.INT
'Now, poor uncle is coming.' (Seki 2000: 61, ex. 21)

¹⁴ Some data in Seki (2000) may, at first sight, contradict the claim that =a occurs with possessed nouns in this language. Thus, several possessive phrases are shown where the possessed noun does not appear with =a:

- (i) a. Kunu'um-a rup.
boy-N father
'Boy's father.' (Seki 2000: 56, ex. 2b)
- b. Kunu'um-a akan
boy-N head
'Boy's head' (Seki 2000: 56, ex. 3b)

One also notes from the source that, in the instances where possessed nouns are presented without -a and translated with adnominal possession, they are presented in isolation, as opposed to being within a sentence. I suspect the possessive relationship encoded by these structures is not the one reflected in the translation of these examples, namely, adnominal possession. That this seems to be the case is supported by instances in which the same structure is translated with predicative possession when presented within a sentence in the source (iia), but receives a translation of adnominal possession when shown in isolation (iib). By contrast, when used within a sentence, the same form pyt always appears with =a and is translated with adnominal possession (iic).

- (ii) a. Po ne = pyt
Q 2SG = house
'Do you have a house?' (Seki 2000: 160, ex. 498)
- b. Je = pyt
1SG = house
'My house' (Seki 2000: 57, ex. 5b)
- c. Je = pyr-a amoete.
1SG = house-N far
'My house is far.' (Seki 2000: 163, ex. 515)

- c. **Na = je = pyr-ite.**
NEG = 1SG = house-NEG
'I do not have a house.' (Seki 2000: 161, ex. 499)
- d. **(Na =)je = 'amor-ite = ne = ko = py.**
(NEG) = 1SG = miss.someone-NEG = ASS = FS = MS
'I do not miss (someone).' (Seki 2000: 330, ex. 1228)

A similar distribution of *=a* is found in Tapirapé (Praça 2007), where it is referred to as the 'referrer' (hence, glossed as REF). Thus, it appears with verbal arguments, complements of postpositions, and possessed nouns. The formative is also required in clausal nominalizations. Note that *=a* does not arise in Tapirapé nouns ending in a vowel, which the source represents with a zero allomorph.

- (48) a. **Miãr-a** mĩ a-yj kã'ã-pe
deer-REF HAB 3.A-run jungle-LOC
'The deer always run around the jungle.' (Praça 2007: 40, ex. 88)
- b. **Takãr-a** yptype xãytãtã'i-ø i-exãk-i xyre-ø = agỹ-ø
takãra-REF near start-REF 3.B-see-AIS boy-REF = PL-REF
'Near the (men's house) takãra, the boys saw the stars.' (Praça 2007: 149, ex. 416)
- c. **Xe = memyr-a** i-xinyk
1SG = son-REF 3-be.sad
'My son is sad.' (Praça 2007: 40, ex. 89)
- d. **Ãpi-ø** a-kwããw [tãtã-ø 'o-pã-ãw-a].
mother-REF 3.A-know banana-REF eat-COMPL-NMLZ-REF
'Mother knows (they) ate all the bananas.' (Praça 2007: 119, ex. 329)

Again, as with Kamaiurá and Ka(g)wahíva, *=a* is absence from vocatives in Tapirapé (49a). What is more, in the absence of this morpheme on forms that otherwise are simple nouns with it, the result distributes akin to a verbal predicate; this is demonstrated in (49b), where negation marking is by means of the standard negator of prototypical verbs (49c).

- (49) a. **Kãtowyg** e-xar ãpy
Kãtowyg 2SG.IMP-come before
'Kãtowyg, come here, please.' (Praça 2007: 43, ex. 97)
- b. **Nã-xe = memyr-i.**
NEG-1SG = offspring-NEG
'I don't have children.' (Praça 2007: 192, ex. 568)
- c. **Kã'ã-pe** mĩ n = a-kãxym-i.
jungle-LOC HAB NEG = 3.A-lose.way-NEG
'In the jungle, he never gets lost.' (Praça 2007: 211, ex. 630)

As shown, the distribution of *=a* is consistent across TG languages. It can be generalized as follows: *=a* satisfies the syntactic requirement that a word must bear it to function as a noun.

Next, I turn to the analysis of this morpheme in Ka(g)wahíva. This will be followed by a review of non-nominalizing approaches to the cognate morpheme in the family in Section 6.4.

6.3 =a, a nominal categorizing head

Assuming, with Halle & Marantz (1993: and following works), that members of lexical categories, like nouns (verbs, and so on), are not atomic elements, but rather uncategorized roots that distribute as nouns by means of syntactic derivation with a functional categorizer ('little' *n*, 'little' *v*), I characterize =a as a nominal categorizing functional head ('little' *n*) in Ka(g)wahíva. That is, this enclitic derives nouns from acategorical roots, or from members of other lexical classes. To show this schematically, in (50), =a categorizes the root as a noun and heads an NP (50), while in (51), =a categorizes an already categorized root (here, as a verb, by *v*) to form a noun.

(50) [_{NP} [_{√root}] [_{n=a}]]

(51) [_{NP} [_{VP} [_v [_{√root}]]]] [_{n=a}]]

I note that, with members of other lexical classes, =a's role is more similar to that of a 'nominalizer' that converts a non-nominal element into a nominal one. Therefore, the traditional sense of 'nominalization' is appropriate to characterize the syntactic derivation involved in the combination of =a with postpositions, adjuncts, and verbs.

Further evidence for =a's nominal categorizing of acategorical roots comes from a non-nominal distribution in the absence of =a. Speaker translates the same word without =a as an existential predicate. For example, in (52a), the form *kapyr* 'leaf, money' functions as a predicate 'to have money'. Distributionally, in these uses, these forms are verbs, as they bear sentence negation, similar to how more typical verbs like 'u 'eat' can be negated, as in (52b). In contrast, the same form *kapyr* cannot appear with sentential negation in combination with =a, as seen in (52c). I take it, by hypothesis, that its verbal distribution means that in these instances, the *a*-less forms have been categorized by a null *v*.

- (52) a. Ndakapyri jie.
 Nd-a-kapyr-i jie.
 NEG-1SG.A-leaf-NEG 1SG
 'I do not have money.' (Elicit)
- b. Nda'ui ki jie pira.
 Nd-a-'u-i ki jie pira.
 NEG-1SG.A-eat-NEG PST 1SG fish
 'I did not eat fish.' (Elicit)
- c. *Ndakapyrai jie.
 Nda-kapyr=a-i jie.
 NEG-leaf=NMLZ-NEG 1SG
 'Intended: I do not have money.' (Elicit)

A reviewer asks if the behavior of *a*-less words like *kapyr* in (52a) is not more indicative of an underlying verb, rather than an acategorical root, in which case =a attaches not to an acategorical root. We will now see that the proposed analysis of acategorical roots, but not the alternative where they are verbal, also makes the right predictions for other phenomena in Ka(g)wahíva. The argument will come from allomorphic and allosemantic patterns involving the nominal past tense and negation morphology with simple nouns, but not derived nominals.

We have seen that Ka(g)wahíva nominals, simple (50) and derived (51), exhibit nominal tense morphology. Importantly, nominal tense, in particular the past tense marker, is lexically conditioned in simple nouns (-*ver* ~-*kwer*) (53a)-(53b), but only -*ver* surfaces in deverbal nominals, as in (53c) and (53d).

- (53) a. Ay-**ver** = a
wood-**NOM.PST** = NMLZ
'Stick.' (Also applicable to any piece of wood that is no longer part of a tree)
(Júma: Text and Elicit)
- b. Ji = rembireko-**kwer** = a = hěa
1SG.POSS = partner-**NOM.PST** = NMLZ = 3SG.FEM
'My ex-wife.' (Common in speech)
- c. Ji = u'u-(a)**ver**-a
1SG.POSS = bite-**NOM.PST**-NMLZ
'My bite.' (Júma: Text)
- d. Kwarupi e-mõmbe'u [ga v-uva = ga repiag-(a)**ver** = a
today 2SG.IMP-tell 3SG.MASC 3SG.COR = father = 3SG.MASC see-**NOM.PST** = NMLZ
oi'i]!
yesterday
'Tell (them) today that he saw his father yesterday.' (Júma: Elicit)

I take it that T being inside = *a* in all nominals, by Mirror Principle reasoning, means that T is closer to the root than *n*. I propose that the T head is acategorial and can combine with a root before it is categorized; (54) updates the structure in (50) for simple nouns when T is present. The structure of deverbal nominals, with T present, is given in (55).

$$(54) \quad [_{nP} \ n=a \ [_{TP} \ T \ [\sqrt{root}]]]$$

$$(55) \quad [_{nP} \ n=a \ [_{TP} \ T \ [_{vP} \ [_{v_0} \ [\sqrt{root}]]]]]$$

Following the intuitions that i) categorizing heads are cyclic/spell-out-inducing heads in the morphology (Moskal 2015; Sande et al. 2020: among many others) and that ii) contextual allomorphy and allophony are constrained by spell-out domains (Marantz 2013: among others), this proposal predicts that we should see freer allomorphy of T with respect to the root in nominal structures than in verbal ones. The lexically conditioned allomorph of the past tense marker in simple nominals is expected, then, if, in nominalized clauses, there is a *v* head that instantiates a domain boundary for allomorphy not present in simple nominals, where T attaches to an uncategorized root.

The negator *-e'ym* also exhibits an idiosyncratic pattern in simple nominals, such as (56), but not derived nominals (57). First, observe that in both instances, Neg comes inside = *a* (and T, for that matter). By the Mirror Principle again, Neg is closer to the root than *n*.

- (56) U'i-**e'ym** = a
 $\sqrt{flour}$ -**NOM.NEG** = NMLZ
'(I ate) flourless (rice).'

- (57) E-mõmbe'u [gã mãjioka erur-**e'ym** = aver = a]!
2SG.IMP-tell 3PL manioc bring-**NOM.NEG** = PST = NMLZ
'Tell (them) that they did not bring manioc!'

Negation, however, is constrained by the choice of root in simple nominals (58); no such restriction is found in nominalized clauses (59):

- (58) * $\sqrt{vyhy}$ -**e'ym** = a kavarapea
 $\sqrt{blood}$ -**NOM.NEG** = NMLZ cockroach
'Intended: Cockroaches do not have blood.'

- (59) E-mõmbe'u [kavarapea $\sqrt{vyhy-\emptyset-e'ym} = a$]
 2SG.IMP-tell cockroach $\sqrt{blood-v-NOM.NEG} = NMLZ$
 'Tell (them) that cockroaches do not have blood.'

The item *vyhy* 'blood' is used in (58) as an element of a nominal predicate; in this simple nominal structure, *-e'ym* is unacceptable (cf. *u'i-e'ym = a* 'flourless' above). In (59), the very same root is used as the predicate of an embedded clause, where (by hypothesis), it has first been verbalized before being negated. As predicted, *-e'ym* cannot select/not select for a particular root in this context because the verbalizer *v* creates an additional domain boundary blocking such selection.¹⁵

On the alternative that roots are underlying verbs without *=a*, these systematic idiosyncrasies involving tense and negation in simple, but not derived nominals, are unexpected. There would be no spellout boundary between roots and tense and negation heads in either type of nominal, and thus, we would expect them to arise across nominals, contrary to truth.

I believe the data presented in this section can also speak favorably to the status of *=a* as realizing *n* in Ka(g)wahíva, rather than realizing some other bit of morphology, as a reviewer suggests. Let us assume that under this alternative, the noun-categorizing head *n* is null. In this case, the null formative would create a spellout boundary between the root and tense and negation across all nominals. If so, then the pattern of allomorphy/allosemy just described would still be surprising, this time because it exists at all, as T and Neg would never have access to the root across *n*.

6.4 Other approaches to *=a* across TG

Although cognates of *=a* are regarded as a nominal formative in all descriptive studies within the family (Rodrigues 1953; Lemos Barbosa 1956; Rodrigues 1996; 2001; Vieira 1995; Queixalós 2006), there have been two main opposing views on its function: a nominalizing vs. a non-nominalizing formative.

The earlier works that considered this morpheme, while working within no particular framework, treat *-a* in the dormant language Tupinambá and in Tapirapé (or label the morpheme) with a nominalizing function (Rodrigues 1953; Almeida et al. 1983; Lemos Barbosa 1956).¹⁶ To my knowledge, the generative-based work by Vieira (1993; 1995), which includes a chapter on NP structure and NP quantification in Asuriní do Trocará, is the only other proposal with this view, and also the most systematic defense of the nominalizing view of *-a* among all.¹⁷

Two other proposals, which are framed in the functionalist tradition, have in common that *-a* is not a nominalizing marker: the case marker and the 'referential' marker. The analysis of *-a* as a case marker comes from Rodrigues (1996; 2001). Based on data from Tupinambá, Rodrigues suggests that Tupinambá *-a* is the 'argumentative' case; he also refers to *-a* as 'nominative' and 'nominal' case; this analysis has been extended to other languages in the family (Seki 2000; Borges 2006; Duarte 2007; Magalhães 2007; Cardoso 2008: to cite just a few). Finally, *-a* is regarded as a 'referential' marker in Queixalós

¹⁵ In simple nominals, negation has the idiosyncratic privative meaning. In the present account, this can similarly be made to follow from negation being closer to the root in simple nominals but not in deverbal nominals, thus allowing for allosemy of negation in this context.

¹⁶ According to Almeida et al. (1983: 31-32), "Nominalizing the verbal root through the nominalization process of nominal roots, that is, adding the suffix *-a* to roots ending in a consonant, the result is 'an action noun', which can be used with verbs expressing perception".

¹⁷ As Vieira (1993: 25) writes: "It is through the suffix *-a* that nouns are derived in Asurini", as well as, "[*-a*] is a syntactic operator whose function is to turn a stem into a noun" (Vieira 1995: 704).

(2006). As a referential marker, *-a* is described as being capable of instituting a referential expression, but does not necessarily have to.

In dos Santos (Under review), I review and compare these approaches as applied to the Ka(g)wahíva dataset. Here, I summarize the main reasons these approaches cannot account for the distribution of *=a*, at least in Ka(g)wahíva.

As far as the case marking analysis is considered, this analysis solely rests on Rodrigues (2001)'s observation that Tupinambá *-a* is in complementary distribution with other morphemes that he also considers realizing Case. In Ka(g)wahíva, however, cognate members of the putative paradigm in Tupinambá may co-occur. Furthermore, Rodrigues (2001)'s alternative labeling of *-a* as 'nominative' leads to the expectation that *-a* should also behave as a nominative case marker. This expectation is not confirmed in Ka(g)wahíva either: while *=a* appears in all nominal elements in a clause, including non-subjects, cross-linguistically, nominative case is tied to subjects (i.e., external argument in unergative and transitive, and internal argument in unaccusatives) and is unique per clause.

As for the 'referential' marking approach, in dos Santos (Under review), I clarify the terminology 'referential', which likely denotes some form of definiteness in this approach. Based on this understanding, I demonstrate that under no sense of a definite marker, based on a typology of definites (Dawson & Jenks 2023: 108-109), is *=a* a definite marker in Ka(g)wahíva.

I have indicated that the non-nominalizing accounts of *-a* (i.e., the case marker and 'referential' marker) are aligned with the functionalist tradition. I do not take that as a coincidence. In fact, it is the very assumption from this framework that lexical categories are inherent that stands between the two opposing views for *-a*. An explicit rejection of the nominalizing status for this morpheme is found in Cruz & Praça (2019: 631), and serves to illustrate the theoretical nature of this dissent: "It is important to observe that the suffix *-a* does not promote lexical class change, and therefore is not a nominalizer, since it occurs with both nouns and verbs."

Since the present work is not committed to the same view that lexical categorization is inherent, but assumes that it arises by means of syntactic derivation (indeed, it is warranted in Ka(g)wahíva given its straightforward account of several patterns of allomorphy and allophony involving tense and negation morphology), the distribution of *=a* across simple and other nominals (e.g., deverbal nouns) is straightforward: *=a* is a root-attaching formative only in the former instances. In the first use, as the root is category-neutral, no lexical change is promoted, just like Cruz & Praça (2019: 631) observed for simple nouns. However, in the second use with e.g., verbs, a change in lexical category is evident; this claim is not disputed in the literature on Tupí-Guaraní, where dependent clauses are viewed as nominalizations that contain an internal verbal core.

So far, we have concluded that Ka(g)wahíva *=a* is the exponent of the nominalizer *n*, and that nouns with *=a* can be the sole elements of a noun phrase that appear in argument position. This distribution makes them bare nouns in argument position. Next, I provide arguments that argumental bare nouns are indeed bare.

6.5 Bare nouns are not covert DPs

The freer distribution of bare nouns in argument position in Ka(g)wahíva is not matched by the behavior of bare nouns in other well-known languages, like English, where only a smaller set of nouns (bare plurals and mass nouns) can occupy an argument position without a determiner.

This raises the oft-explored question of the structure of bare nouns. Some argue that, as nominal arguments are necessarily DPs, bare nouns are always DPs (Longobardi 1994); by hypothesis, languages without overt articles involve a covert DP. Others argue that bare nouns are NPs (Chierchia 1998; 2009), or are ambiguous between NP and DP (Franks & Pereltsvaig 2004), an example of which is the Algonquian language Inuu-aimun (Gillon 2011).

Against this backdrop, probing the structure of Ka(g)wahíva bare nouns becomes a relevant issue given the claim that complement clauses are nominalized TP-embeddings under a *nP*, lacking a DP. However, if a covert DP is available in some languages, it could, in principle, be available for Ka(g)wahíva as well.

To address this issue, in this section I draw on semantic diagnostics to defend that simple nouns with *=a* are bare *nPs*, and not a covert DP. The same diagnostics also show that Ka(g)wahíva bare nouns do not vacillate between *nP* and DP structures, as Gillon (2011) has shown for bare nouns in Inuu-aimun. Table 3 summarizes the diagnostics and the results expected under each approach to bare nouns laid out above, including the results for a language with an ambiguous *nP* and DP. The rightmost column, then, includes the results for Ka(g)wahíva under each diagnostic. As one can see, a covert DP structure is not available for simple nominals. Consequently, it must then not be available for derived nominals, like nominalized complement clauses, either.

	<i>nP</i>	DP	Innu-aimun	Ka(g)wahíva
wide scope/escape scope	✗	✓	✓	✗
narrow scope	✓	✗	✓	✓
uniqueness effects	✗	✓	✗	✗
can be used anaphorically	✗	✓	✓	✗
must be used anaphorically (in stretches of discourse)	✗	✓	✗	✗
obeys law of contradiction	✗	✓	✓/✗	✗
generic and kind readings	✗	✓	✗	✗

Table 3: Ka(g)wahíva bare *nPs* are not covert DPs.

Bare nouns overwhelmingly take narrow scope. In pragmatically neutral contexts, which are associated with VSO clauses, bare nouns have a narrow scope reading. We see this in (60).

- (60) Ojuka Tampinha-ga karuvoruhua ra'e, 'i ki ore kora.
 O-juka Tampinha = ga **karuvoruhu** = a ra'e, 'i ki ore ko = ra.
 3.A-go T. = 3SG.MASC **paca** = NMLZ TODAY.PST say PST 1EXCL REALIS = TODAY.PST
 'Tampinha killed (a) paca, we said.'
 Context: The speaker directly quotes what she and others said as they saw
 Tampinha arrive with a paca. (Júma: Text)

Example (61) provides additional textual confirmation for the narrow scope interpretation associated with bare nouns. The second sentence states the outcome of an agouti-hunting outing, and it also reveals that the bare noun *iramburava* 'agouti' in the first sentence has a narrow scope reading.

- (61) Ore hoi iramburava reka orokovo, 'itekiko
 Ore ho-i iramburav = a rek = a oro-ko-vo, 'i:te:ki:ko
 1.EXCL go-AIS **agouti** = NMLZ look.for = NMLZ 1.EXCL.COR-be-DEPEND DIR.EVID
 rōmña. Ndorepiaki ki ore ko.
 rōmña. Nd-or-epiak-i ki ore ko.
 FRUST NEG-1.EXCL-see-NEG PST 1.EXCL REALIS
 'We had looked for an agouti. We did not see (any).' (Júma: Text)

A narrow scope reading is the only alternative available with propositional attitude verbs like *pota* 'want, desire' (62a), in conditional clauses (62b), and in questions (62c).¹⁸

- (62) a. Opotaete ga tapy'ynha kwarupi.
 O-pota-ete ga **tapy'ynh = a** kwarupi.
 3.A-want-REALLY 3SG.MASC **non.indigenous** = NMLZ today
 ✓ Context 1: 'He wants (to date) any non-indigenous person today.'
 ✗ Context 2: 'He wants (to date) a specific non-indigenous person today.'
 (Júma: Elicit)
- b. Agui-ga oje'o javatxinga ekiramẽ] nãhẽ.
 Agui = ga o-je'o [**javatxing = a** ekir = a = mẽ] nãhẽ.
 Agui = 3SG.MASC 3.A-cry **dog** = NMLZ enter = NMLZ = IF POT
 ✓ Context 1: 'Agui will cry if any dog enters (the house).'
 ✗ Context 2: 'Agui will cry if a specific dog enters (the house).' (Júma: Elicit)
- c. O'u po ndera'yra-ga pira?
 O-'u po nde = ra'yra = ga **pira = a**?
 3.A-eat IRR 2SG.POSS = offspring = 3SG.MASC **fish** = NMLZ
 ✓ Context 1: 'Does your son eat any fish?'
 ✗ Context 2: 'Does your son eat a particular fish?' (Júma: Elicit)

Another diagnostic for the presence of a covert D is uniqueness. If Ka(g)wahíva bare nouns had a covert definite DP, then we would expect them to be associated with uniqueness effects, that is, they must refer to a unique referent within a context. (Note that, as uniqueness is associated with a definite determiner, the lack of a uniqueness effect is not evidence for a lack of DP structure. Therefore, this diagnostic only tests whether bare nouns are built as definite DPs.

Ka(g)wahíva texts show that bare nouns lack uniqueness effects. Consider the following excerpts from the same text, in which the speaker narrates an eventful weekend in a different village. Among the activities was going hunting *karuvoruhua* 'paca'. The text has five occurrences of *karuvoruhu* = a, (63a)-(63e), which cannot be a unique referent.

¹⁸ With negation, bare nouns can scope under negation, as in (i), a pattern that is commonly found for bare nouns crosslinguistically (Deal & Nee 2018). Additionally, dos Santos (2024b: 69-70) had previously shown that Ka(g)wahíva bare nouns also outscope negation. Since this behavior is exceptionally found only with negation, I set it aside here.

- (i) Ndoropyhygite ore pira, 'i ki gã ko.
 Nd-oro-pyhyg-i-te ore **pira** = a 'i ki gã ko.
 NEG-1.EXCL.A-catch-NEG-really 1.EXCL **fish** = NMLZ say PST 3.PL REALIS
 ✓ Context 1: 'They caught no fish.'
 ✗ Context 2: 'There is a particular fish that they did not catch.'

- (63) a. Aramẽ txiro ore, txaho kavyripe rehe, karuvoruhua rehe,
 Aramẽ txiro ore, txa-ho kavyr = ipe at **karuvoruhu** = a rehe,
 AFTER.THAT PROGR. 1.EXCL OPT-go jungle = to at **paca** = NMLZ at
 ore 'ei, 'itekiko nõmã.
 ore 'e-i, 'i:te:ki:ko nõmã.
 1.EXCL say-AIS DIR.EVID FRUSTR
 'After that, we go to the jungle after pacas, we said.' (Line 72)
- b. Karuvoruhua txaho epiaga ore, 'i ki ka kora.
karuvoruhu = a txa-ho epiag = a ore, 'i ki ka ko = ra.
paca = NMLZ OPT-go see = NMLZ 1.EXCL say PST 3SG.MASC REALIS = TODAY.PST
 'We go to see pacas, he said.' (Line 80)
- c. Karuvoruhua!, 'i ki ka kora.
karuvoruhu = a, 'i ki ka ko = ra.
paca = NMLZ say PST 3SG.MASC REALIS = TODAY.PST
 'A paca!, he said.' (Line 88) Context: Said as someone in the group spots something under the water which, to him, appeared to be a paca.' (Line 88)
- d. Karuvoruhua txia jive ko nõmã, 'i ki ka
karuvoruhu = a txi = a ji = ve ko nõmã, 'i ki ka
paca = NMLZ that = NMLZ 1.SG = to REALIS FRUST say PST 3SG.MASC
 kora.
 ko = ra.
 REALIS = TODAY.PST
 'To me, it was a paca, he said.' (Line 96)
- e. Ojuka Tampinha-ga karuvoruhua ra'e, 'i ki ore kora.
 O-juka Tampinha = ga **karuvoruhu** = a ra'e, 'i ki ore ko = ra.
 3.A-go T. = 3SG.MASC **paca** = NMLZ TODAY.PST say PST 1.EXCL REALIS = TODAY.PST
 'Tampinha killed a paca, we said' (Line 114)

Anaphoricity can also be used to probe the structure of bare nouns. If bare nouns occur with a covert determiner, we would expect all uses of a bare noun in a stretch of discourse to refer to the same entity. We have just seen that this is not true: the several occurrences of *karuvoruhu* = a in (63) do not pick out the same referent.

In fact, the strategy for subsequent reference to an entity previously introduced by a bare noun is simply not pronouncing that noun again. In the same text from which (63e) from above draws upon, the referent of the last occurrence of *karuvoruhu* = a in (63e) is referred back to in (64). As seen, no overt noun is used.¹⁹

- (64) Kwari-ga rekoi, ipiroga, 'itekiko.
 Kwari = ga reko-i, i-pirog = a, 'i:te:ki:ko.
 Kwari = 3SG.MASC be.with-AIS MVD.OBJ-peel = NMLZ DIR.EVID
 '(Then), Kwari stayed peeling (the paca).' (Line 115)

¹⁹ Additionally, the verb is obligatorily marked with a prefix *i-*. In Ka(g)wahíva, the pattern of *i*-prefixation is triggered just in case the complement of a head is moved or dropped; thus, it is found with verbs, postpositions, and inalienable possessed nouns. Despite the initial impression that examples like (64) may give, this morpheme is not a resumptive pronoun, if we consider that resumptive pronouns draw from the inventory of a language's pronouns (McCloskey 2017: 4), while *i-* is not part of the pronominal inventory of Ka(g)wahíva.

Furthermore, the pattern of *i*-prefixation is widespread across the Tupí-Guaraní family. The literature on these languages regards *i-* as a marking of syntactic adjacency between a complement and its selecting head (Rodrigues & Cabral 2012).

In the rare instances where a bare noun matches the description of a previously introduced NP, the reference is disjoint. The three sentences in (65) are uttered one right after the other. The first sentence has *pira* ‘fish’, as well as the second one, but neither refers to the same referent. In the third sentence, the referent of *pira* from (65b) is referred back; as expected, no overt NP is used.

- (65) a. Karuka = mē ki ji ho-i **pira** = a reru = a oi’i.
 afternoon = to PST 1.SG go-AIS **fish** = NMLZ bring = NMLZ some.time.ago
 ‘In the afternoon, I went to bring fish (by fishing).’
 b. **Pira** = a rerur-i.
fish = NMLZ bring-AIS
 ‘(I) brought (a) fish.’ (Juma: Text)
 c. Ko’ema = mē jie i-mōtyryry-i
 morning = to 1.SG MVD.OBJ-fry-AIS
 ‘On the next morning, I fried (the fish).’ (Juma: Text)

Additionally, bare nouns with =a do not pass the ‘test of consistency’ or ‘law of contradiction’ (Löbner 2002). According to this test, a noun referring to an individual cannot both hold and not hold a given property.

- (66) a. Tapi’ira ehe. Tapi’ira ndahei.
Tapi’ir = a ehe. **Tapi’ir** = a nda-he-i.
tapir = NMLZ be.tasty **tapir** = NMLZ NEG-be.tasty-NEG
 ‘There is tapir that is delicious, there is tapir that is not delicious.’ (Júma: Elicit)
 b. Kunhã ikatu. Kunhã ndikatui.
Kunh = ã i-katu. **Kunh** = ã nd-i-katu-i.
woman = NMLZ MVD.OBJ-be.pretty **woman** = NMLZ NEG-MVD.OBJ-be.pretty-NEG
 ‘There is one woman who is pretty, and there is a woman who is not pretty.’
 (Júma: Elicit)

Finally, bare nouns are also used for generic and kind readings.²⁰

- (67) a. Ndavyi kavarapea.
 Nda-vy-i **kavarape** = a.
 NEG-blood-NEG **cockroach** = NMLZ
 ‘Cockroaches do not have blood.’ (Júma: Elicit)
 b. Mbahira vapo kunhã.
 Mbahira v-apo **kunh** = ã.
 Mbahira 3.A-make **woman** = NMLZ
 ‘Mbahira created women.’ (Jupaú: Text)

In sum, bare nouns in Ka(g)wahíva exhibit the semantic profile of bare *n*Ps, and not the profile of a noun with a covert DP structure.

Next, I present data showing that a Ka(g)wahíva simple nouns may still project the category D through a semantically restrictive set of determiners. This discussion comple-

²⁰ Note that the word for blood is pronounced as vy in (67a), in contrast to vyhy in the earlier example in (58). The surface form vy is the result of two regular phonological processes in Júma: /h/ deletion between identical vowels in an unstressed syllable (here, the syllable /hy/ becomes unstressed due to stress shift driven by the negative suffix -i), followed by vowel hiatus resolution, after the former process creates a vowel sequence.

ments the background on nominals, which I will draw upon to assess the nominal status of complement clauses.

6.6 Determiners and demonstratives

In Ka(g)wahíva, the elements that are commonly analyzed as members of the determiner category are (in)definiteness markers and demonstratives.

Starting with the definite enclitics, the full set of enclitics is presented in Table 4. In the ethnolects under discussion here (Júma and Jupaú), these enclitics impose a strict semantic restriction on their complement nouns – they combine only with nouns whose referents are living humans or God.²¹

DETERMINER	GLOSS
= <i>ga</i>	‘SG.MASC definite determiner’
= <i>hẽa</i>	‘SG.FEM definite determiner’
= <i>gã</i>	‘PL definite determiner’
= <i>mõ</i>	‘indefinite determiner’

Table 4: (In)definiteness determiners.

These enclitics exhibit the profile of definite determiners. For instance, nominals marked with them cannot appear in existential contexts, as the pairs with and without the definite enclitics illustrate below.²²

- (68) a. Ndokoi akwemba’ea jahya pyri.
 Nd-o-ko-i akwemba’e = a jahya pyri.
 NEG-3.A-be-NEG man = NMLZ moon near
 ‘There is no man on the moon.’ (Júma: Elicit)

²¹ Furthermore, nouns in combination with any of the definite enclitics always exhibit =*a*. Relatedly, as examples from Section §6.1 already showed, neither =*a* nor the definite enclitics appear in the vocative.

A reviewer suggests this distribution of proper names in the vocative may lend substance to the idea that =*a* is an overt D-related element. The idea draws on a potential parallel to Greek vocatives, in which vocatives do not exhibit a determiner Stavrou (2014). As a result, Stavrou treats vocatives as structurally smaller than a DP. In turn, the reviewer suggests a parallel between Greek and Ka(g)wahíva vocatives: the latter’s vocatives are smaller than a DP as well, and since =*a* is absent in the vocative, the absent material is a D-head.

I believe this parallel would be less than a coincidence had Ka(g)wahíva vocatives also lack overt material that is otherwise required. In particular, as I already pointed in this note, not only =*a* is absent in the vocative, but the definite enclitics discussed in the present section cannot appear in the vocative either.

Following closely the suggestion above, pairing the assumption that Ka(g)wahíva vocatives are structurally smaller to the absence of =*a* does not ultimately require that =*a* be a D-head.

Indeed, in the view that =*a* is a categorizing *n* in simple nouns and the definite enclitics are determiners, as I argue, the structural parallel with Greek vocatives could still be partially maintained. Specifically, Ka(g)wahíva vocatives are structurally smaller, lacking the enclitics, but that does not automatically mean that =*a* is the D-head.

Furthermore, the idea defended here that =*a* is a *n*-head and the definite enclitics realize a D-head is harmonic with the semantic behavior of nominals, as described by the several pieces of evidence put together in Section §6.5 and the present section. In contrast, if =*a* were a D-head, the striking *n*P-behavior of bare nouns would be mystifying.

²² Other diagnostics are not available in Ka(g)wahíva, given the restriction that these determiners combine with nouns whose referents are living humans. Therefore, it is not possible to use them to describe parts of previously-introduced wholes (e.g., I bought a bicycle for my son, but the tires were worn out).

- b. *Ndokoi akwemba'ea-**ga/gã** jahya pyri.
 Nd-o-ko-i akwemba'e = a = **ga/gã** jahya pyri.
 NEG-3.A-be-NEG man = NMLZ = **3SG.MASC/3.PL** moon near
 Intended: 'There is no man on the moon.' (Júma: Elicit)
- (69) a. Ndokoi 21mẽ tapy'ynha.
 Ndo-ko-i 21 = mẽ tapy'ynh = a.
 NEG-3.A-be-NEG 21.village = to non.indigenous = NMLZ
 'There are no non-indigenous people in the 21 (village).' (Júma: Elicit)
- b. *Ndokoi 21mẽ tapy'ynha-**gã**.
 Ndo-ko-i 21 = mẽ tapy'ynh = a = **gã**.
 NEG-3.A-be-NEG 21.village = to non.indigenous = NMLZ = **3.PL**
 Intended: 'There are no non-indigenous people in the 21 (village).' (Júma: Elicit)

The enclitic determiner = *ga* '3rd singular masculine' is used to refer to a unique referent like *Tupan* 'God'.

- (70) Mārānũhũ po txia txirove? Tupana-garõi auhu.
 Mārān-ũhũ po txia txirove? Tupan = a = **ga** = rõi auhu.
 why-AUG IRR DEM now God = NMLZ = **3SG.MASC** = CONST.NEG seem
 'Why (is) this now? It seems it is not God.' (Lamenting about the COVID situation)
 (Júma: Text)

Nouns, in combination with the enclitic determiners, pass the 'test of consistency' or 'law of contradiction' (Löbner 2002). This is expected, as definite determiners create arguments that refer to individuals, and definite nouns obey the law of contradiction. For example, in (71), *kunhã* 'woman' must refer to the same individual in each instance. Therefore, the enclitic must be a definite determiner.

- (71) #Kunhã-**hẽa** ikatu. Kunhã-**hẽa** ndikatui.
 Kunh = ã = **hẽa** i-katu. Kunh = ã = **hẽa** nd-i-katu-i.
 woman = NMLZ = **3SG.FEM** MVD.OBJ-be.pretty woman = NMLZ = **3SG.FEM** NEG-MVD.OBJ-be.pretty-
 Intended: 'The woman is pretty, and the woman is not pretty.' (Júma: Elicit)

Ka(g)wahíva also exhibits a nonspecific indefinite determiner = *mõ* in contexts where the speaker does not have a particular referent in mind, as in (72).

- (72) Ndahepiagi ki ji piramõ.
 Nd-a-hepiag-i ki ji pira = a = **mõ**.
 NEG-1SG.A-see-NEG PST 1SG fish = NMLZ = **INDEF.DET**
 'I did not see any fish.' (Said upon returning from a fishing trip with no fish)
 (Júma: Text)

Nouns in combination with = *mõ* exhibit the typical behavior of non-specific indefinites: they must take narrow scope relative to negation and verbs like *pota* 'want, desire' and within conditional clauses; see dos Santos (2024b: 70-71) for examples.

Finally, I present the demonstratives on account that they may instantiate the category D in Ka(g)wahíva. Thus, demonstratives show a three-way deictic distinction, which distinguishes between proximal, medial, and distal. Each of these, in turn, shows a two- or three-way categorization of their referent, distinguishing between elongated, round, and flat. This system yields the rich paradigm presented in Table 5.

Table 5: Demonstratives.

	elongated	round	flat
proximal	<i>‘āngā</i>	<i>koro</i>	<i>kohoa</i>
medial	<i>mīhīa</i>	<i>txiro</i>	<i>pehea</i>
distal	<i>mīhīa</i>	<i>txijia</i>	<i>txijia</i>
examples	mortar	ball, puppy, tapir	hose

Demonstratives follow the noun (73a), but they can also appear to its left and discontinuous from the NP when focused (73b).

- (73) a. Javatxinga txijia
 Javatxing = a **txijia**
 puppy/dog = NMLZ **DEM:DIST:ROUND/FLAT**
 ‘That (distant) puppy’ (Júma: Elicit)
- b. Txijia ki javatxinga omõnhãn tapi’ira ko.
 Txijia ki **javatxing = a** o-mõnhãn tapi’ira ko.
 DEM:DIST:ROUND/FLAT PST **dog = NMLZ** 3.A-corral tapir REALIS
 ‘That dog corralled the tapir.’ (Júma: Elicit)

Taken altogether, the data presented in this subsection show that some Ka(g)wahíva nouns may project a DP. However, excluding the indefinite enclitic = *mõ*, the other determiners are very selective of the semantics of their noun complements. Therefore, if there is any chance a nominal can exhibit a D, it is likely to be with = *mõ*.

To summarize this background section, Ka(g)wahíva nominals may be bare nouns, the combination of an acategorial root and the formative = *a*, or a noun with one of the best candidates for a D-head in the language, the (in)definiteness markers, or the demonstratives.

Given these results and the hypothesis that clausal complements are nominalizations, it should be straightforward to see whether they also project a DP: they should be able to appear with one of the Ka(g)wahíva D-elements.

I turn now to the nominal properties of complement clauses to examine this possibility as well.

7 Nominal-like properties of complement clauses

Clausal complements exhibit several nominal properties. Syntactically, they occur in object position, as several of the examples illustrated thus far show; example (74a) is given for convenience. They also occur in (derived) subject positions (74b)-c).

- (74) a. Ahepiang ki jie jiruva-ga pira ‘ua.
 A-hepiang ki jie [ji = ruva = ga pira ‘u = a].
 1SG.A-see PST 1SG 1SG.POSS = father = 3SG.MASC fish eat = NMLZ
 ‘I saw [my father eating fish]’ (Júma: Elicit)
- b. Ahy ki ndererekwavera oi’i.
 Ahy ki [nde = rereko-(a)ver = a] oi’i.
 be.hard PST 2SG.B = be.with-NOM.PST = NMLZ some.time.ago.
 ‘Caring for you is hard.’ (Júma: Elicit)

- c. Ahy moja jiu'uavera.
 Ahy [moja ji = u'u-(a)ver = a].
 be.hard snake 1SG.B = bite-NOM.PST = NMLZ
 'That the snake bit me hurts.' (Júma: Elicit)

Additional evidence for their nominal status draws on their morphosyntax. First, observe that these structures require the nominal-exclusive enclitic = *a*, which I have proposed to be the overt exponence of the *nP* head.

- (75) Ahepiang ki jie jiruva-ga pira 'ua.
 A-hepiang ki jie [ji = ruva = ga pira 'u = *(a)].
 1SG.A-see PST 1SG 1SG.POSS = father = 3SG.MASC fish eat = NMLZ
 'I saw my father eating fish.' (Júma: Elicit)

Perhaps unsurprisingly, the nominal tense and negation morphology is also common to complement clauses.

- (76) Tense is nominal in complement clauses
- a. Garembiriko-ver = a = hẽa jie arekoajam.
 Ga = rembiriko-**ver** = a = hẽa jie a-reko-ajam.
 3SG.MASC = partner-NOM.FUT = NMLZ = 3SG.FEM 1SG 1SG.A-marry-again
 'His ex-wife, I married (her).' (Júma: Elicit)
- b. Emõmbe'u Aguia-ga mbiara mbohirukaravera javatxinga pe!
 E-mõmbe'u Aguia = ga mbiara mbohirukar-**aver** = a javatxinga pe!
 2SG.IMP-tell A. = 3SG.MASC meat drop-NOM.PST = NMLZ dog to
 'Tell (them) that Agui dropped the meat for the dog!' (Júma: Elicit)
- (77) Negation is nominal in complement clauses
- a. Ere'u nde mbotavoruhua.jauhuva'ea u'ie'yma rehe?
 Ere-'u nde mbotavoruhua.jauhuva'ea u'i-**e'ym** = a rehe?
 2SG.A-eat 2SG rice flour-NOM.NEG = NMLZ at
 'Do you eat rice without flour? (Júma: Elicit)
- b. Emõmbe'u gã mǎjioka rerure'ymavera!
 E-mõmbe'u [gã mǎjioka rerur-**e'ym**-(a)ver = a]!
 2SG.IMP-tell 3.PL manioc bring-NOM.NEG-NOM.PST-NMLZ
 'Tell (them) that they did not bring manioc!' (Júma: Elicit)

These results strongly suggest that Ka(g)wahíva does project a nominal structure in clausal nominalizations, and that this structure contains at least *nP*, whose head is spelled out by = *a*.

Does Ka(g)wahíva also project the crosslinguistically more common nominal layers in clausal nominalizations, that is, DP and PossP? In the remainder of this section, I argue that neither projection is afforded in these structures.

Starting with PossP, one piece of evidence that other works have relied on to argue for its presence in clausal nominalizations is the case expression of subjects. For instance, in Khalkha Mongolian, subjects in nominalized complement clauses may appear in the genitive form, similar to possessive nominals, as in examples (78a)-b).

- (78) a. Pronoun possessor in genitive form in Khalkha
 Chi **minii** nom-iig unsh-san
 2SG.NOM 1SG.GEN book-ACC read-PST
 'You read my book' (Lim 2026: 6)

- b. Nonfinite nominalized complement clause with genitive subject in Khalkha
 Naraa [**minii** unta-j bai-sn]-iig(= min') med-ne.
 N.NOM **1SG.GEN** sleep-PROG COP-PFV-ACC = 1SG.POSS know-N.PST
 'Naraa knows that I was sleeping' (Lim 2026: 34)

Likewise, subjects in Turkish are marked genitive in TP nominalizations, but not in other types of embeddings; contrast (79)-(80).

- (79) Nominative subject in finite embeddings in Turkish
 Hasan [**uşak** oda-yı temizle-sin] isti-yor.
 Hasan **servant.NOM** room-ACC clean-OPT/SBJNCT.(3.SG) want-PR.PRG.(3.SG)
 'Hasan wants that the servant should clean the room.' (Kornfilt & Whitman 2011: 1301)
- (80) Genitive subject in nominalized embedding in Turkish
 Hasan [**uşağ-ın** oda-yı temizle-me-sin]-i isti-yor.
 Hasan **servant-GEN** room-ACC clean-NF.NMLZ-(3.SG)-ACC want-PST.(3.SG)
 'Hasan wants that the servant should clean the room.' (Kornfilt & Whitman 2011: 1301)

For Kalkha (Lim 2026), on the assumption that the genitive case is assigned by the nominal structure PossP, the author argues that the expression of genitive subjects in nominalized clauses suggests the projection of PossP.

The diagnostic from genitive subjects is not available in Ka(g)wahíva – pronominal possessors are not formally distinct from regular pronouns that occur in argument positions, whether they are in a matrix or in an embedded clause. Examples (81) readily illustrate this isomorphism. The same data, however, reveals a prosodic difference: the first person singular pronoun is a proclitic in (81a), but a free-standing word in (81c).^{23,24,25}

- (81) a. Epiang jikyva!
 E-piang **ji** = kyva!
 2SG.IMP-see **1SG** = lice
 'See my lice!' (Júma: Elicit)
- b. Amōka'ē ji e'yi pira.
 A-mōka'ē **ji** e'yi pira.
 1SG.A-roast **1SG** MANY.EX fish
 'I roasted many fish.' (Júma: Elicit)
- c. Okwaha ki ga ji nderepiaka.
 O-kwaha ki ga [**ji** nde = repiak = a].
 3.A-know PST 3SG.MASC **1SG** 2SG.B = see = NMLZ
 'He knew I saw you.' (Júma: Elicit)

²³ I thank a reviewer for drawing my attention to this prosodic distinction between the pronominal forms.

Incidentally, a variation in the pronunciation of the first person singular *jia* subject as subjects in clauses, but not in possessor phrases, might also be suggestive that the first person singular is treated differently in these two contexts. As subjects, the first person is sometimes pronounced as [ɕʲe], [ɕe], or [ɕi]. In contrast, such variation is never found in possessive phrases with the first person singular, where only [ɕi] is heard. Like procliticization, we can see this difference as stemming from the distinct syntactic structures in which the pronoun is used.

²⁴ Pronominal forms notoriously surface as proclitics when occurring as the object of postpositions, the object of verbs, and inalienable possessors across Tupí-Guaraní languages (Jensen 1990; 1998; Meira & Drude 2013; Rose 2015; dos Santos 2023).

²⁵ That *ji* in (81b) is a standalone word, rather than the possessor of fish, receives confirmation from the intervening *e'yi*, which is an adjunct.

(82) $[_{VP} \nu [_{NP} n=a [_{TP} T [_{VoiceP} FA [_{Voice} [_{VP} [_{\nu} [_{IA} \sqrt{root}]]]]]]]]]$

(83) a. Mǎngǎ nde erepiang piauhua ipyhyga?
Mǎngǎ nde ere-piang [__ piauhua i-pyhyg = a]?
who 2SG 2SG.A-see piauhua.fish MVD.OBJ-catch = NMLZ
 ‘Who did you see catching the piauhua fish?’ (Júma: Elicit)

b. Mǎngǎ pira ka txiro omõngi, a’i ki ji kora.
Mǎngǎ pira ka txiro o-mõngi __, (a-’i ki ji ko = ra).
who clothes 3SG.MASC PROGR 3.A-dress 1SG.A-say PST 1SG DIR.EVID
 ‘Whose clothes is he dressing in? (I asked).’ (Júma: Text)

c. *Mǎngǎ ka txiro omõngi pira?
Mǎngǎ ka txiro o-mõngi [__ pira]?
who 3SG.MASC PROGR 3.A-dress clothes
 Intended: ‘Whose clothes is he dressing?’ (Júma: Elicit)

Note that this distribution directly contrasts with the pattern in English gerundive nominalizations, where both possessors and genitive subjects behave alike under extraction: neither can be extracted.

- With that, I take this Ka(g)wahíva pattern, in which embedded subjects, but not possessors, can be extracted, to be highly suggestive that they occur in different structural positions. Otherwise, we would not be able to explain the asymmetry in extraction. This fact also

A second related question regarding clausal nominalizations concerns the relationship between nominalizer type and the amount of structure they can nominalize. Recent work has attempted to identify crosslinguistic restrictions on the amount of content that a given nominalizer may embed (Alexiadou 2020; Iordăchioaia 2020; Lim 2025; 2026). An upshot of these discussions for the present discussion is that high layers of clausal structure, such as CP or TP, may be nominalized only by a high layer of nominal structure, such as DP (or PossP for TP-embedding nominalizations in Lim (2025; 2026)). On the other hand, low levels of clause structure (below TP) are in principle compatible with both levels of nominal structure (cf. Alexiadou (2020: 102), who suggests that DPs cannot merge directly with ν Ps, and it needs n P before taking a ν P). Particularly relevant to us is their claim that n -nominalizations should not be able to embed TPs, but should allow a structure smaller than a TP.

Given the results arrived at for Ka(g)wahíva in this paper, these theories are too restrictive. I have shown that Ka(g)wahíva complement clauses are TP-embedding nominalizations by n P, not DP. In fact, no DP is projected in them. This pattern is simply unexpected under those proposals.²⁶ Providing an alternative explanation as to why Ka(g)wahíva allows TP-sized nominalizations by n P will take us too far in this paper. This, I leave for future research.

For now, I should note that the proposals in Alexiadou (2020) and Lim (2026: footnotes 35, 36) allow some flexibility in their account, which may be explored for a potential account for the Ka(g)wahíva pattern. In particular, the authors consider the cut-off points for nominalization – how much clausal structure content can be nominalized by what – not as driven by an absolute cut-off, but as based on a partitioning of the clausal structure in domains that are semantically-grounded (into discourse linking, anchoring, point of view, classification) (Ramchand & Svenonius 2014; Wiltschko 2014). In this view, nominalizations target these semantic domains: for instance, because discourse linking is commonly attributed to CP in the clausal structure and to DP in the nominal domain, D-nominalization, but not n -nominalization, can target CP structures. In contrast, the domain classification is part of ν P in the clausal structure, and to n P in the nominal structure; therefore, n -nominalizations may target ν P for nominalization.

Under this idea, we could say that in Ka(g)wahíva, discourse linking is realized by CPs and DPs, while anchoring is realized by TPs and n Ps. Initial support for the idea that n Ps pertain to the anchoring domain comes from the fact that bare nouns (bare n Ps) are allowed in argument position. Being on the same domain allows n to target a TP for nominalization.²⁷

²⁶ A parallel pattern seems to be in Oshiwambo, as discussed in Lee & Ndapo (2025). They argue that TPs are the topmost layer in subject nominalizations in the Bantu language Oshiwambo, which they assume is a n P-nominalization. However, they also include a representation for these nominalizations on p. 14 that contains a richer nominal structure than that proposed for Ka(g)wahíva, where a DP is projected on top of the n P projection that directly embeds the TP. They point out that this structure poses a challenge to Alexiadou (2020)'s proposal that rules out TP-nominalization by n . As far as I can tell, it does not challenge Lim (2025; 2026)'s, which predicts TP-embedding nominalizations by nominal structure that also includes a DP.

²⁷ A referee is concerned about the idea that Ka(g)wahíva bare nouns are n Ps and that they can also be related to an anchoring domain – much like TPs are considered as part of an anchoring domain (Ramchand & Svenonius 2014; Wiltschko 2014) – unless "bare nouns do indeed signal some reference".

For anchoring, the important property is having a time parameter, which clausal tense do. Do bare nouns in Ka(g)wahíva exhibit a time parameter? While going into the details to answer this empirical question is out of the scope of the paper, I can initially offer the observation that the presence of nominal tense is suggestive of a time parameter in these nouns. In particular, since nominal tense is understood as semantically similar to clausal tense in languages like the sister language Mbyá (Thomas 2014), and Ka(g)wahíva bare nouns

Finally, by the same idea that nominalization targets semantically-grounded domains, why then does Ka(g)wahíva not allow CP nominalizations by D, since the language exhibits elements that may project a DP layer with the (in)definiteness enclitics and demonstratives? The only firm answer is the one I suggested in section 7. That is, the absence of D in clausal nominalizations is likely due to the restrictor *nP*, at least in the instances with definite enclitics and demonstratives. That is, clause-sized *nPs* cannot serve as a restrictor *nP* because they do not refer to living humans (or God), nor can they be categorized by shape, as demonstratives require from their *nPs*.²⁸ However, a similar semantic restriction is apparently not in place for the indefinite enclitic = *mõ*. Nevertheless, it is simply the case that a clausal nominalization with the indefinite is not possible (see Example (89)).²⁹

9 Conclusion

In this paper, I have set out to investigate the structure of clausal nominalization through Ka(g)wahíva, focusing in particular on nominalized complement clauses. A common assumption in works about clausal nominalization is that they consist of a nominal layer on top of the clausal content, which exhibits verbal properties (the Mixed Extended Projection / Nominalization Functional Thesis) (Borsley & Kornfilt 2000; Kornfilt & Whitman 2011). A question that emerges in this literature is what exactly the nominal layer in clausal nominalizations looks like. Another related question concerns the amount of clausal content a given nominalizer may embed. In exploring Ka(g)wahíva nominalized complement clauses in detail, I have argued that their nominal layer simply consists of *nP*, with no further nominal projections (e.g., a DP). Furthermore, complement clauses are TPs nominalized by *nP*. The first result stands in contrast to proposals for other languages, which exhibit clausal nominalization by D, while the second result indicates that TP nominalization by *nin* is possible, a pattern not predicted in the recent literature that seeks to relate nominalizer type to the size of the embedded structure. Each result has only very recently been achieved for Japanese and Khalkha (Mongolic) (nominalizations by *nP*) (Nasu et al. 2024; Lim 2025), and Oshiwambo (Bantu) (TP-embedded nominalizations by *nP*) (Lee & Ndapo 2025).

Abbreviations

A = subject agreement, ACC = accusative, AIS = adjunct-initial suffix, ASS = assertive, ATEN = attenuative, ATT = attested, B = object agreement, CIRCUMST = circumstantial, COMPL = completive aspect, CONST.NEG = constituent negation, COP = copula, COR = coreferential marking, DEM:MED:ROUND = demonstrative for referents at a medium distance from the speaker and having a round shape, DEM:DIST:ROUND = demonstrative for referents at a distance from the speaker and having a round shape, DEPEND =

exhibit nominal tense cognates with Mbyá (dos Santos 2024b: 57-58), then Ka(g)wahíva nominal tense is the best indication for a time parameter in these nouns.

²⁸ Interestingly, the semantic restriction imposed by the determiner onto its restrictor, as observed in Ka(g)wahíva, is not observed with the D elements that nominalize clauses in languages that realize the D-assumption of clausal nominalization, where the determiner simply turns predicates into individuals, without asking whether the NP referent is animate or has a particular shape.

²⁹ If it turns out that this combination is also ruled out on semantics grounds, then a defender of the D-assumption of clausal nominalization could say that the Ka(g)wahíva facts do not actually affect the soundness of their theory because the presence of all D elements is ruled out on semantic grounds, not syntactic ones. I thank a reviewer for raising this point.

dependency marker, DIR.EVID = direct evidential marker, DISTANT.PST = distant past, FOC = focus, FRUSTR = frustrative, FS = final sentence particle, GEN = genitive, HAB = habitual aspect, INDF.DET = indefinite determiner, IRR = irrealis mood, LOC = locative, MANY.EX = exact quantifier, MVD.OBJ = moved (or dropped) object, MS = male speaker, N = nominal, NF.NMLZ = non-factive nominalization, N.INT = non-interrogative, NMLZ = nominalizer, NOM = nominative, NOM.NEG = nominal negation, NOM.PST = nominal past, N.PST = non-past, N.WIT.PST = non-witnessed past, OBJ = object, OPT = optative, PFV = perfective, POT = potential, PROGR = progressive, PROSP = prospective, Q = interrogative, REALIS = realis mood, RED = reduplicant, REF = referrer, SBJNCT = subjunctive, VIS = visual, VOC = vocative

Ethics and consent

Uncited data was obtained through field research that has been approved by the Institutional Review Board at the University of California-Berkeley (number 2019-04-12071) and the Museu dos Povos Indígenas/FUNAI.

Funding information

Field research that led to the uncited data reported in the paper has been made possible due to the following funding opportunities: several iterations of the Oswalt Grant for Field Research (UC Berkeley), a grant from the Programa de Documentação de Línguas Indígenas (Prodoclin) (Museu dos Povos Indígenas/FUNAI), a Language Legacies Grant (Endangered Language Fund), and a Glottobank small grant (Endangered Language Documentation Programme).

Acknowledgements / Agradecimentos

Minha incomensurável gratidão aos povos Ka(g)wahiva cujas línguas estão representadas aqui, em especial àqueles que contribuíram diretamente para este trabalho. Em especial, agradeço a minha mestra de Juma, Mandéi Juma, e aos meus mestres de Jupaú (Uru Eu Wau Wau), Awip (Davi) e Puré Uru Eu Wau Wau, que participaram das sessões de elicitación. Também agradeço a Mandá Uru Eu Wau Wau, da aldeia Jaikara/621, que se disponibilizou a narrar histórias e relatos pessoais cujos excertos também fazem parte do conjunto de dados Jupaú apresentado neste trabalho.

I also thank Peter Jenks for feedback on this work. Additionally, I would like to thank the Editor David Erschler and the anonymous reviewers for their instigating comments that made this paper better. Errors are my own.

Competing interests

The author has no competing interests to declare.

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